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20 Electrical Stimulation

David Levine and Barbara Bockstahler

Terminology Used in Electrical Stimulation

Electrical stimulation (ES) is a commonly used modality in physical therapy, which is effective for many purposes, including increasing muscle strength, muscle reeducation, increasing range of motion (ROM), correction of structural abnormalities, improving muscle tone, enhancing function, pain control, accelerating wound healing, edema reduction, muscle spasm reduction, and enhancing transdermal administration of medication (iontophoresis).¹ Terminology associated with electrical stimulation can be confusing, and, in 1990, a committee of the electrophysiologic section of the American Physical Therapy Association (APTA) developed the *Standards of Electrotherapeutic Terminology*, a document created to unify and standardize the terms and definitions used by biomedical engineers, researchers, educators, and clinicians. Historically, electrical stimulators and the terminology related to them have been referred to by their specific rehabilitative application, the use of the inventor's name, or the commercial companies that produced them. Examples include galvanic current, faradic current, diadynamic current, high voltage, low voltage, low frequency, medium frequency, transcutaneous electrical nerve stimulation (TENS), electrical muscle stimulation (EMS), functional electrical stimulation (FES), Russian stimulation, and interferential stimulation. Unfortunately the multitude of names has created confusion regarding the physiologic effects and clinical results. Transcutaneous electrical nerve stimulation has been widely used to identify stimulators that modify pain, whereas neuromuscular electrical stimulation (NMES) or electrical muscle stimulation (EMS) has been identified with muscle reeducation, prevention of muscle atrophy, and enhanced joint movement. Accurate terminology dictates that almost all electrical stimulators are TENS units, as they work transcutaneously through surface electrodes to excite nerves. In the typical scenario where the muscle is innervated by a motor nerve, NMES is the appropriate terminology, and when a muscle is denervated and requires direct muscle fiber activation through electrical stimulation, the term EMS is used.

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Neuromuscular electrical stimulation, the primary focus of this chapter, is a form of clinical electrotherapy used to treat a wide variety of physiologic disorders or injuries in humans and is recognized as a promising treatment modality for similar disorders in veterinary patients. By definition, NMES is the administration of an electrical current generated by a stimulator that travels through leads to electrodes placed on the skin to depolarize the motor nerve and produce a skeletal muscle contraction. Stimulation of motor end plates with electrical current causes nerve depolarization and subsequent activation of muscle fibers.^{2,3}

This chapter describes the myriad of potential clinical uses of electrical NMES; therefore this term is used throughout. In addition to the discussion of general uses of NMES, specific attention is paid to ruptured anterior cruciate ligament (ACL)/cranial cruciate ligament (CrCL) rehabilitation because of the relatively high incidence in humans and dogs and the amount of research information that is currently available on the use of NMES for knee/stifle rehabilitation. The emerging use of electrical stimulation to treat pain in the dog is also examined.

History of Electrical Stimulation

Electricity has been used therapeutically since Scribonius Largus used an electrical ray in a footbath to treat gout. Benjamin Franklin used electrical shock to treat a frozen shoulder in 1757.³ Electrical muscle stimulation has been specifically used in physical therapy since the mid-1700s, when electrostatic generators were used to treat patients with paralysis. In 1791 Galvani used galvanic current in *in vitro* experiments using neuromuscular preparations. By the early 1800s Faraday had invented the faradic current generator, which is the basis for most modern muscle stimulators. Documentation of NMES to reduce the loss of muscle weight and prevent atrophy in denervated muscles occurred in the early to mid-twentieth century. Electrotherapy has since become commonplace in physical medicine for the restoration of muscle function after injuries, before patients are capable of voluntary exercise training.⁴

Basic Concepts of Electrical Stimulation

To understand the benefits that electrical stimulation may provide in veterinary practice, a basic explanation and definition of the electrical current parameters used are summarized in [Box 20-1](#).

Electrical Current/Waveforms

Three types of currents are commonly used: (1) continuous direct current (DC) ([Figure 20-1, A](#)), (2) continuous alternating current (AC) ([Figure 20-1, B](#)), and (3) pulsed current (either AC or DC) ([Figure 20-1, C](#)). Continuous direct current is a unidirectional electrical current that flows for 1 second or longer; alternating current changes the direction of flow at least once every second. Continuous direct current (also called galvanic current) has been used for wound healing, iontophoresis, and in the treatment of denervated muscle. In current clinical practice it is used only for iontophoresis. Continuous alternating current is not used therapeutically, but is our typical line current (see [Figure 20-1, B](#)).

Pulsed current consists of a unidirectional or bidirectional flow of charges (DC or AC) that periodically stops

Box 20-1 Typical Parameters Available in Neuromuscular Electrical Stimulation Devices

Waveform: The shape of the visual representation of pulsed current on a current/time plot or voltage/time plot. Can be symmetric, asymmetric, balanced, unbalanced, biphasic, monophasic, polyphasic, and so forth.

Amplitude: The current value in a monophasic pulse or for any single phase of a biphasic pulse.

Phase/pulse duration: The duration of a phase or a pulse, usually measured in microseconds.

Pulse rate or frequency: The rate of oscillation in cycles per second, expressed as pulses/sec (pps) or hertz (Hz). Often labeled as pulse rate or pulses/sec, or frequency on stimulators.

On/off time: The amount of time the stimulator is delivering current compared with the rest period between contractions, usually measured in seconds.

Ramp: The time in seconds from when the current begins to the peak current (example: 3-second ramp up, 6-second contraction, 2-second ramp down).

Polarity: Electrode may be either the anode (+) or cathode (−). (This is not relevant when using AC.)

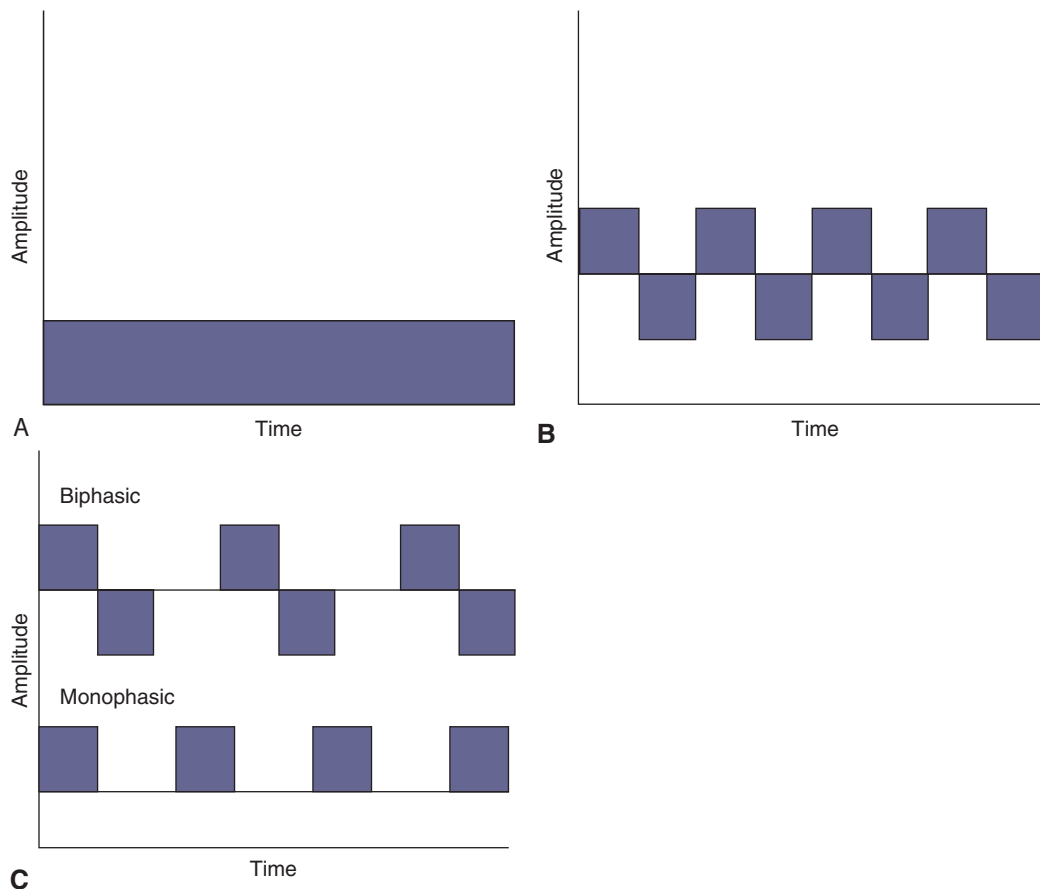


Figure 20-1 A, Continuous direct current. B, Continuous alternating current. C, Pulsed currents (AC and DC).

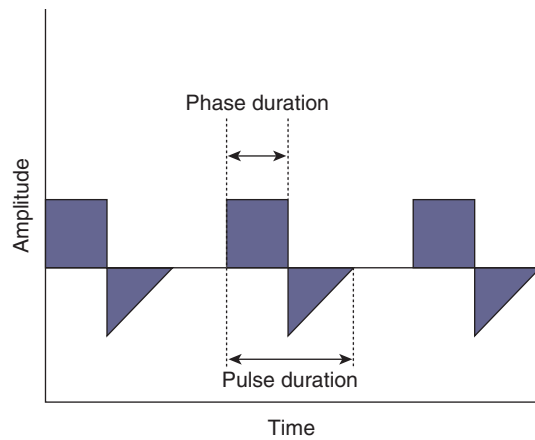


Figure 20-2 Phase and pulse duration.

for a finite time period. All NMES devices are pulsed current stimulators. Phase duration (also referred to as phase width) is defined as the time in which the current flows from the baseline in one direction and back to the baseline (Figure 20-2).

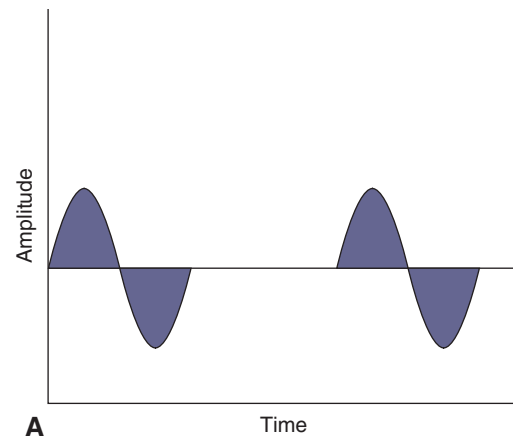
Pulse duration (also referred to as *pulse width*) is defined as the time during which charge flows in both directions (see Figure 20-2). In a monophasic current, the pulse duration and the phase duration are the same. In a biphasic current, two phases make up one pulse. Pulsed current consisting of a bidirectional flow of charge is called biphasic pulsed current.

When the flow in each direction is the same in time and amplitude-dependent features, the biphasic current pulse is considered *symmetric* (Figure 20-3, A). If the time- and amplitude-dependent features differ, then the biphasic current is termed *asymmetric* (Figure 20-3, B). Balanced asymmetric biphasic pulsed current occurs when the total charge in one phase equals the total charge of the other phase (Figure 20-4). If the total charge in each phase is unequal, it is termed *unbalanced* (Figure 20-5). Both pulsed AC and DC current forms are commonly used in portable and clinical model NMES units.

Stimulator Parameters

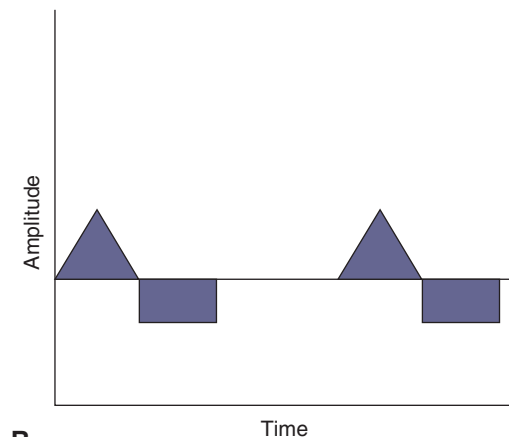
Current amplitude (also called magnitude or intensity) is defined as the vertical distance from the highest to the lowest peak during one electrical wave and is typically measured in milliamperes (mA) (Figure 20-6). Increasing the amplitude induces a stronger force of muscle contraction by recruiting additional muscle fibers at greater distances from the electrodes.^{1,3}

Skin produces resistance to current flow by ohmic resistance and capacitive impedance. Capacitive impedance is patient dependent and cannot be modified. Clipping or shaving the hair coat and cleaning the skin with alcohol to wash away skin oils or other substances helps to lower ohmic resistance. Lowering the skin resistance diminishes the driving voltage that is necessary for current penetration



A

Time



B

Figure 20-3 A, Symmetric biphasic current. B, Asymmetric biphasic current.

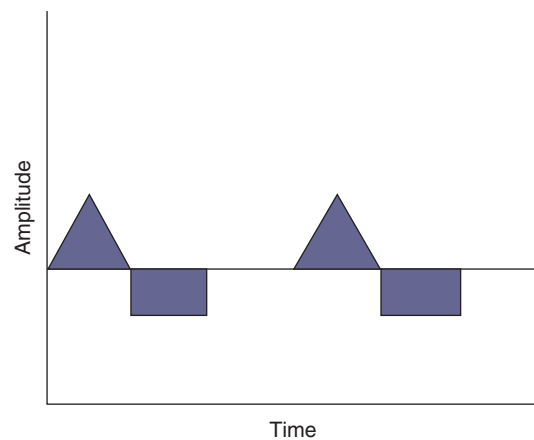


Figure 20-4 Balanced asymmetric biphasic pulsed current.

of the skin, potentially making the treatment more comfortable.^{1,3}

Increased current amplitude is required to produce a given amount of muscle contraction force if the pulse or phase durations are short (<100 μ sec). Portable NMES units are not always capable of producing the current

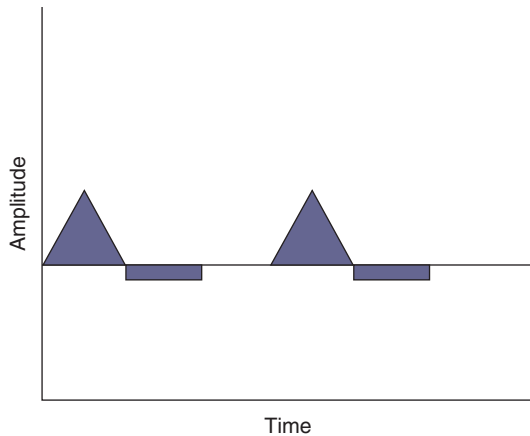


Figure 20-5 Unbalanced asymmetric biphasic pulsed current.

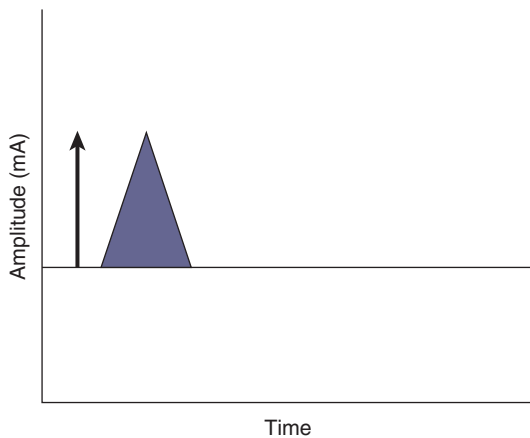


Figure 20-6 Amplitude.

output that clinical models can produce; however, most units sold for electromedical purposes are adequate. Some bargain simulators possess a very low phase charge (<10 microcoulombs) and cannot create adequate contraction of large muscle groups.

Pulse durations of 150 to 400 microseconds produce powerful contractions while minimizing the likelihood of recruiting many pain fibers.^{1,3} Some stimulators that are made only for NMES do not allow control of pulse duration, but are typically set in this range. As the pulse duration increases, smaller diameter pain fibers are recruited. One study has examined pulse durations in dogs and subsequently recommended pulse durations by individual muscles (Table 20-1). This study also examined the differences in needed pulse durations between people and dogs. Although they are similar, dogs in general do not require as long a pulse duration as human motor nerves to depolarize. This can potentially increase comfort by using the shortest recommended pulse duration.⁵

Pulse rate (also called frequency, rate and pulses per second [pps]) is the number of pulses delivered per second and is measured in hertz (Hz). Tetanic muscle contractions

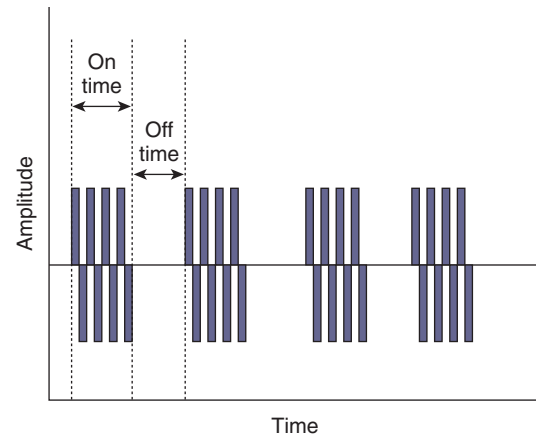


Figure 20-7 Duty cycle.

may be produced with frequencies as low as 20 Hz, but only submaximal muscle contractions are typically produced in this range. Maximal force of contraction generally occurs between 60 and 100 Hz. However, as frequency increases the rate of fatigue also increases. Lower tetanic frequencies in the range of 35 to 50 pps reduce muscle fatigue while still providing strong muscle contractions.^{1,3}

Duty cycle is the ratio of on-time to total cycle time, expressed as a percentage (Figure 20-7). On-time is the period of time in which a series of pulses or bursts are delivered to the patient. Off-time is the time between on-times. A single on-time duration plus a single off-time duration constitutes the total cycle time. For example, a stimulator that causes a muscle to contract for 10 seconds followed by a 30-second rest would have a 25% duty cycle, or a 1:4 ratio. As on-time increases, muscle fatigue increases.² Optimal duty cycles vary, depending on the patient. A patient with severe atrophy may require a longer off-time to recover between contractions. Many clinicians start with duty cycle ratios between 1:2 and 1:5 and watch for signs of fatigue, which indicates the need for a longer off-time.

Ramp is a feature of NMES that helps improve patient comfort. It involves a gradual increase or decrease in current amplitude such that the number of recruited motor units gradually increases the force of muscle contraction or gradually decreases the force of contraction (Figure 20-8). The ramp time is the period of time over which the pulse is increasing or decreasing. No data to date have identified the optimal ramp time. However, a ramp up time of 2 to 4 seconds is commonly used to maximize comfort with a 1- to 2-second ramp down.¹ Suggested current parameters for muscle strengthening are listed in Box 20-2.

Electrodes

Many types of surface electrodes are available. The main criteria in choosing electrodes are as follows:

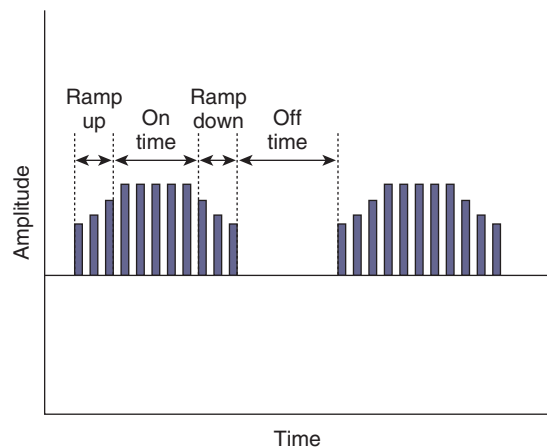
1. Flexible enough to conform to the tissue
2. May be trimmed to a specific size

Table 20-1 Recommended Pulse Durations for 11 Studied Muscles in Dogs

Muscle	Chronaxy Value (μ s)	Range of Recommended Pulse Durations for Use in Dogs* (μ s)	Reported Human Chronaxy Range (μ s)	Range of Recommended Pulse Durations for Use in Humans (μ s)
Supraspinatus	167 $\pm$ 55 (133-200)	150-190	80-160	100-300, 150-250, 250
Infraspinatus	151 $\pm$ 61 (113-188)	130-170	80-160	100-300, 150-250, 250
Deltoides	180 $\pm$ 60 (138-221)	160-200	80-160	100-300, 150-250, 200
Triceps brachii (lateral head)	217 $\pm$ 74 (171-262)	200-240	160-320	100-300, 150-250, 200
Extensor carpi radialis	190 $\pm$ 69 (141-238)	170-210	200-360	300, 150-250, 200
Gluteus medius	179 $\pm$ 69 (136-221)	160-200	80-160	300, 250-400, 400
Biceps femoris	207 $\pm$ 63 (167-246)	180-220	440-720	300, 250-400, 400
Semitendinosus	166 $\pm$ 25 (150-181)	150-190	440-720	300, 250-400, 400
Vastus lateralis	233 $\pm$ 73 (181-282)	210-250	80-160	300, 250-400, 400
Tibialis cranialis	238 $\pm$ 111 (160-314)	210-250	200-360	300, 250-400, 400
Erector spinae	205 $\pm$ 79 (155-255)	180-220	160-320	250, 150-250, 300

*These recommended pulse durations are intervals centered over the mean values calculated in this study with an approximate range of $\pm 10\%$ or $\pm 20 \mu$ s. This range facilitates the selection of a single chronaxy value when stimulating several muscles in 1 region.

From Sawaya SG, Combet D, Chanoit G, Thiébaud JJ, Levine D, Marcellin-Little DJ: Assessment of impulse duration thresholds for electrical muscle stimulation (chronaxy) in dogs. *Am J Vet Res* 69:1305-1309, 2008.

**Figure 20-8** Ramp.

Box 20-2 Current Parameters for Strengthening

Frequency: Generally between 25 and 50 Hz. (These have been shown in humans to produce strong tetanic contractions while minimizing fatigue.) An alternative in dogs that are not very tolerant of electrical stimulation is to use a low frequency (<10 Hz). This will provide a twitch contraction and is better tolerated in some cases.

Waveforms: Many waveforms exist; any waveform capable of depolarizing the muscle is acceptable.

Pulse or phase duration: Between 150 and 250 microseconds.⁶⁹

Optimal pulse durations maybe found in [Table 20-1](#).

Ramp up/down (rise and decay time): Adjust between 2 to 4 seconds up to increase comfort, 1 to 2 seconds down.

On/off time: A 1:3 or 1:4 ratio; 10 seconds on, 30 or 40 seconds off is commonly used. This may be decreased as muscle strength improves. A 1:1 or 1:2 ratio is usually used for muscle endurance training.

3. Have a low resistance
4. Are highly conductive
5. May be used many times
6. Are inexpensive

Some commercially available electrodes are effective for only a few uses, and some may be used 100 or more times (e.g., carbon-impregnated silicon rubber electrodes, metal, etc.). Conductive performance of any electrode decreases over time. Electrodes require a medium to transmit current. Commonly used media include gels, moistened sponges or paper towels; some electrodes have the media already applied. Sponges and paper towels tend to dry out, and rewetting is necessary every 30 minutes. Electrodes should be of the appropriate size to stimulate the desired muscle without stimulating unwanted muscles in close proximity. The smaller the electrode, the higher the current density that enters the muscle, and the more uncomfortable the stimulus may be.

An alternative to traditional electrodes is the use of electrodes with multiple fine wires or prongs that make contact with the skin (see Fig. 20-13). The advantage of these types of electrodes is that the hair does not need to be clipped. Rather the fine wire prongs are worked down among the hair to contact the skin. Wetting the hair before treatment helps with conductivity.

Recruitment

Neuromuscular electrical stimulators recruit Type II (fast twitch) fibers first, then Type I (slow twitch), which is the reverse of the muscle recruitment pattern in a volitional contraction.⁶⁻⁸ Increasing the pulse duration increases the recruitment of smaller diameter motor units at the same depth, but increasing the pulse duration too much (800 μ sec) may stimulate undesirable fibers (small-diameter pain fibers). Increasing either the amplitude or the pulse duration affects the strength of contraction because additional muscle fibers are recruited. Increasing the frequency results in the existing motor units firing at a faster rate and will increase the strength of contraction, but it also causes more rapid fatigue. Application of NMES at an optimal frequency results in an optimal physiologic response (the desired contraction) while minimizing fatigue. In a healthy uninjured individual, a maximal voluntary muscle contraction produces a greater torque (more powerful contraction) than occurs in an electrically induced contraction. However, patients with injuries or immediately following surgery may be unable or unwilling to produce a maximum voluntary muscle contraction. In these patients NMES may produce a stronger muscle contraction.

What to Look for When Buying a Neuromuscular Electrical Stimulator Unit

An acceptable NMES unit should allow flexibility in a number of parameters, including frequency (adjustable from at least 1 to 50 Hz), adjustable pulse duration (50 to

200 μ sec or more), on/off times, ramp, and multiple channels (at least two to allow co-contraction or alternating contraction of two different muscle groups). Acceptable clinical units may cost \$300 or more.

Indications and Contraindications for Neuromuscular Electrical Stimulation

Neuromuscular electrical stimulation is commonly used in the rehabilitation of human patients who have had orthopedic or neurologic injury. A few examples are patients recovering from fracture repair, ACL reconstruction, and meniscal debridement or repair. Patients with neurologic conditions, such as cerebrovascular accidents, closed head injuries, spinal cord injuries, or other neurologic disease involving paralysis or paresis, may also benefit from NMES. NMES has been used to increase joint mobility, decrease joint contracture, decrease edema, enhance circulation, minimize disuse atrophy, improve muscle strength, retard loss of volitional control, improve sensory awareness, decrease spasticity, diminish pain, and correct gait abnormalities.⁹⁻¹²

In human patients with low thoracic spinal cord injury, functional NMES has been used to produce knee extension with locking of the knees to allow weight bearing. Within 3 weeks of spinal cord injury, up to 50% of quadriceps muscle loss may occur. In one study NMES returned quadriceps muscle mass to near normal.¹¹ In a patient with an incomplete quadriplegic spinal cord injury NMES was used to strengthen a paretic hand. Two weeks of NMES produced a 33% increase in muscle force with no loss of strength 4 weeks after the treatment was discontinued.¹¹ Children with mild cerebral palsy treated with NMES had statistically significant improvements in gross motor, locomotor, and receipt/propulsion skills.¹⁰ Neuromuscular electrical stimulation has also been used on the tibial muscles of children with Duchenne and Becker muscular dystrophy. Neuromuscular electrical stimulation resulted in mild, short-term increases in muscle strength, but it did not alter muscle fatigue. Whether chronic stimulation produces beneficial long-term effects has yet to be determined.¹²

In addition to the clinical use of NMES in people with neurologic and orthopedic disorders, it has been used in denervated muscle to retard atrophy and improve recovery after reinnervation in rats with surgically severed peroneal nerves. The use of NMES in improving reinnervation recovery in other species warrants further study because it may be useful to those with denervation injuries.¹³

There are a number of circumstances in which electrical stimulation should not be used (contraindications), and other circumstances in which electrical stimulation should be used only with caution (precautions). Box 20-3 outlines these contraindications and precautions.

Box 20-3 Contraindications and Precautions for the Use of Electrical Stimulation

Contraindications

High-intensity stimulation directly over the heart
 In animals with pacemakers
 In animals with seizure disorders
 Over areas of thrombosis or thrombophlebitis
 Over infected areas or neoplasms
 Over the carotid sinus
 Any time active motion is contraindicated
 Over the trunk during pregnancy

Precautions

In areas with impaired sensation
 In areas of skin irritation or damage
 Near electronic sensing devices such as ECG monitors
 (possible interference)

Effects of Neuromuscular Electrical Stimulation in the Rehabilitation of Patients with Reconstructed Anterior Cruciate Ligaments

The specific physical effects of NMES on muscle enzymes, fibers, and perfusion has been examined experimentally in human patients with ACL injuries. The effects of NMES on dogs recovering from CrCL rupture are also beginning to be investigated.

Effect on Muscle Enzymes

In a study regarding the effects of low-frequency electrical stimulation on the metabolic profile of skeletal muscle, a portable NMES unit was used to stimulate the knee extensor muscles of sedentary men and women 6 days a week for 6 weeks.¹⁴ No changes in creatine kinase concentrations were found in vastus lateralis muscle samples before or after NMES. There was a small decrease in phosphofructokinase, the rate-limiting enzyme of glycolysis in human muscle, which was only statistically significant in men. Hexokinase, a regulatory enzyme of skeletal muscle glucose phosphorylation, was only significantly increased in the vastus lateralis of women. Krebs' cycle and electron transfer chain marker enzyme activity of fatty acid oxidation increased significantly in both sexes, with the greatest change occurring in women. The authors concluded that NMES can significantly increase the skeletal muscle aerobic-oxidative potential of sedentary subjects. In a study by Arvidsson et al.¹⁵ there was no difference between a postoperative ACL reconstruction–NMES-treated group and an isometric exercise group in the activity of the oxidative enzyme citrate synthase or phosphofructokinase.

Research by Wigerstad-Lossing et al.¹⁶ showed no significant reduction in the activity of the oxidative and glycolytic enzymes citrate synthase and triphosphate dehydrogenase in NMES patients following knee ligament surgery compared with untreated control knee ligament surgery patients, which had decreased enzyme activity.

Effects on Muscle Fibers

The muscle fiber atrophy that occurs with disuse following cruciate injury and surgery is significant. Neuromuscular electrical stimulation may be more effective than volitional exercise in preventing atrophy and overcoming the effects of reflex inhibition in the first few days to weeks after surgery. Electrical stimulation also recruits Type II muscle fibers first, thereby retarding their atrophy to a greater degree than Type I muscle fibers.^{6,7} As Type II fiber augmentation occurs the force of contraction increases, and therefore, strength increases. With volitional contraction in humans a selective or preferential activation of fast-twitch motor units has not been demonstrated. Slow-twitch motor units with Type I skeletal muscle fibers are recruited first, followed by the fast-twitch motor units with Type II skeletal muscle fibers being recruited as the demand for force increases.⁷

Neuromuscular electrical stimulation of the triceps brachialis muscle in monkeys did not affect the distribution of muscle fiber types but enhanced oxidative capacity and caused an increase in fiber size.¹⁷ There was a preferential recruitment of large motor axons that innervate fast-twitch fibers. However, fast-twitch motor units were not exclusively activated because there was also activation of slow-twitch motor units, but to a lesser extent. Motoneurons and muscle fibers were recruited primarily in the superficial portion of the muscle. The morphologic and biochemical adaptations that occurred with NMES were similar to those occurring after voluntary heavy-resistance exercises.

Effect on Perfusion

Neuromuscular electrical stimulation also induced capillary proliferation in monkeys in response to increased muscle blood flow. Because blood flow increased in capillaries around Type II fibers to a greater degree than around Type I fibers, there was a stimulus for proliferation of endothelial cells.¹⁷

Transcutaneous electrical stimulation was used on rat muscle to demonstrate that microvascular perfusion depends on evoked muscle contractions. This finding suggests that NMES should be applied at intensities that produce muscle contraction so that microvascular perfusion of stimulated skeletal muscle is enhanced.¹⁸

Overall Effects of Neuromuscular Electrical Stimulation

The net physical effects of NMES are an increase in muscle strength, muscle mass, and oxidative capacity.^{6,14,16,17,19,20}

Another benefit of NMES is the ability to overcome the effects of reflex inhibition on the quadriceps muscles, and the potential enhanced effect of NMES on subsequent voluntary use of previously electrically activated motor units.⁶ The negative effects of NMES include muscle fatigue and possibly increased pain. Muscle fatigue is the decrease in force-generating ability of a muscle because of recent activity. The rate of muscle fatigue during NMES is much greater than that which occurs during volitional exercises. The reverse recruitment order induced with NMES probably contributes to increased fatigue. However, stimulation intensity, stimulation frequency, and duty cycle may be altered to reduce muscle fatigue.²¹ Increased current levels may recruit pain fibers and result in painful muscle contractions. In addition, discomfort is also associated with psychological factors in humans. Pain may be controlled by decreasing the current intensity to a level that elicits comfortable muscle contractions.²²

Efficacy and Clinical Trials of Neuromuscular Electrical Stimulation for the Anterior Cruciate Ligament Reconstructed Knee

Currier and Kellogg¹⁹ used NMES to treat seven patients after ACL reconstruction. Treatment successfully reduced thigh girth loss compared with three control patients during the first 6 weeks.

Snyder-Mackler et al.⁶ studied the effects of clinical NMES on thigh muscle strength and gait in 10 patients undergoing ACL reconstructive surgery. Patients were randomly assigned to NMES and volitional exercise or volitional exercise alone for 4 weeks. Values obtained for cadence, walking velocity, stance time on the affected limb, flexion excursion of the knee during stance, thigh circumference, and extension and flexion torques were improved in patients receiving NMES. The knees of the patients receiving NMES were stronger in the eighth postoperative week than reported averages for non-NMES patients years after ACL reconstructive surgery. The authors concluded that NMES not only increased muscle strength but also improved functional use of the muscles at least in the immediate postoperative period. The increased muscle strength likely results in partial compensation for the loss of ACL stability.

DeLitto et al.²⁰ compared clinical NMES and isometric contraction with voluntary exercise and isometric contractions in patients who had undergone ACL reconstructive surgery. Each group consisted of 10 patients. Neuromuscular electrical stimulation patients received treatment for 3 weeks during the first 6 postoperative weeks. After their respective treatment period, bilateral maximal isometric measurements of gravity-corrected knee extension and flexion torque were obtained. Greater extension and flexion torque and higher individual thigh-muscle strength gains were reported when NMES was prescribed early in postoperative rehabilitation.

Lieber et al.²³ compared maximal voluntary thigh muscle contraction after 4 weeks of treatment in 20 NMES patients and 20 voluntary contraction patients who had undergone ACL reconstruction within 2 to 6 weeks of the study. They compared the two groups under conditions in which the total muscle tension in both groups was matched. When exercise intensity was matched between the two groups, the magnitude and time course of strength gains were identical. Given the variable intensity of maximal voluntary contraction that NMES patients tolerate, variability in outcomes may be expected between NMES strengthening and voluntary contraction groups. The authors concluded that when strong muscle contractions can be elicited with NMES, significant strengthening is possible.

Wigerstad-Lossing et al.¹⁶ compared NMES using a portable unit and voluntary muscle contractions to voluntary muscle contractions alone in patients immobilized in a cast following ACL surgery. Thirteen patients were randomly assigned to the treatment group and 10 to the control group. Electrical stimulation was applied to the quadriceps muscles only, through windows in the cast, and was initiated on the second postoperative day and continued until the sixth week after surgery. Preoperative data were compared with data obtained in the sixth postoperative week, and the authors concluded that NMES in combination with simultaneously performed voluntary muscle contractions can limit muscle weakness, muscle wasting, and reduction in oxidative and glycolytic muscle enzyme activity during immobilization after knee ligament surgery.

In another study 18 male and 20 female patients undergoing ACL reconstruction were randomly assigned to one of two groups.¹⁵ Individuals in an NMES group received a cast, isometric muscle training, and NMES with a portable unit. Individuals in the control group received a cast and isometric training alone. Transcutaneous electrical stimulation of the quadriceps was initiated during the first postoperative week and continued for a total of 5.5 weeks. Quadriceps muscle wasting was assessed with computed tomography (CT) before and 6 weeks after surgery. Percutaneous muscle biopsies were obtained from the vastus lateralis muscle before surgery, and 1 and 6 weeks after surgery. Electrical stimulation significantly reduced the amount of vastus medialis muscle wasting in female patients as measured by CT, but there was no significant difference in males. Muscle fiber area revealed that cross-sectional area of individual fibers decreased less in NMES patients than in controls, but the difference was not significant. The authors concluded that transcutaneous electrical stimulation was effective in significantly reducing vastus medialis atrophy as measured on CT, but had no significant effect on the vastus lateralis.

The results of clinical trials using NMES for postoperative rehabilitation of the ACL-deficient knee in people generally have demonstrated increased thigh girth, increased

thigh muscle strength, improved cadence, improved walking velocity and stance time, and increased oxidative and glycolytic muscle enzyme activity.

Rehabilitation Programs for Treatment of Anterior Cruciate Ligament Deficiency

In human surgery, rehabilitation of patients after ACL reconstruction is necessary for an optimal outcome.²⁴⁻²⁷ Controversy exists regarding what constitutes the ideal postoperative rehabilitation program for human patients. Current trends in volitional exercise regimens are geared toward accelerated rehabilitation programs. Typical accelerated programs involve ROM exercises and cryotherapy started immediately postoperatively, and crutch use for 2 to 10 days. Closed kinetic chain exercises in the form of exercises in which the foot is fixed and motion at the knee joint is accompanied by motion at the hip and ankle joint (such as squats) are initiated as early as 7 days postoperatively. Closed kinetic chain exercises appear to decrease knee joint compressive forces, thereby decreasing the anteroposterior translation of the tibia, as compared with open-chain knee extension exercises.²⁸ Neuromuscular electrical stimulation is commonly used during the first 3 to 6 weeks of rehabilitation, depending on the ability of the patient to volitionally contract the quadriceps. At approximately week 10 light jogging, Stairmaster, and agility drills may generally be attempted. From weeks 12 to 24 strengthening programs, proprioceptive training, and increased running and agility programs are added as tolerated. Advancement from one level of activity to another is based on physical examination, knee laxity testing, and isokinetic and functional testing.^{16,25,28-30}

Postoperative rehabilitation and the role of the quadriceps and hamstring muscles in improving stifle function have been largely ignored in veterinary medicine. Postoperative care of dogs, regardless of the surgical technique used, traditionally has included several days to weeks of external support to reduce postoperative swelling and provide joint immobilization. This is done to protect the surgical repair until pericapsular fibrosis occurs in the case of extracapsular repair, provide protection of an intracapsular graft while it remodels, protect the site of fixation of the graft, and permit healing of the osteotomy and repair site with a tibial plateau leveling osteotomy (TPLO).³¹⁻⁴² However, immobilization exacerbates muscle atrophy and weakness and produces deleterious effects on cartilage, bone, and ligaments.^{4,15,16,24,26}

Veterinary surgeons should not make the error of using all aspects of the accelerated volitional rehabilitation protocols designed for human patients in their veterinary patients post-CrCL stabilization. The majority of human ACL surgical repairs involve using the bone–patellar tendon–bone technique, which is strongest the day of surgery and loses strength as tendon remodeling occurs. Many canine CrCL-deficient stifles are stabilized using the

extracapsular lateral fabella suture technique, which relies on collagen formation and scar tissue as the primary stabilizer of the joints.⁴³ This technique results in immediate strength following surgery because of the stabilization sutures, and the periarticular structures gain strength as tissue healing and collagen maturation occur over time. Intracapsular techniques typically are not as strong, and the graft weakens over several weeks before regaining some strength. Therefore rehabilitation protocols for canine patients must be based on the surgical stabilization technique used. However, NMES may be useful in the rehabilitation of the stifle without compromising the stability of the repair.

The negative effects of quadriceps reflex inhibition leading to quadriceps atrophy following ACL injury and the positive protective effects of hamstring muscle strengthening to enhance stability and constraint of the stifle following CrCL injury have not been adequately addressed in veterinary medicine. Clinicians may wish to consider adapting ACL-protective rehabilitation procedures to the dog, to protect the surgical repair. Improved postoperative functional outcome following ACL reconstruction in humans is positively correlated with return of quadriceps strength.^{6,19,26} Although the hamstring muscles also atrophy following ACL injury and surgery, the atrophy is generally less than that found in the quadriceps muscles, probably because the hamstring muscles cross both the knee and hip joints, which results in continued function and muscle contraction, thereby diminishing the degree of atrophy.²⁶

Neuromuscular Electrical Stimulation Use and Clinical Recommendations in the Dog

Neuromuscular Electrical Stimulation Use in Veterinary Medicine

In establishing a rehabilitation program, the cost-benefit ratio must be assessed. For a program to be cost and labor effective there must be earlier or more protected return to function, increased joint ROM, increased muscle strength, and an overall improved outcome.⁶ Although cost is a major concern in veterinary medicine, of equal importance is how a rehabilitation program is applied to the dog. Obviously the use of crutches or other limb assistive devices are not directly applicable to the dog. However, external coaptation with soft padded bandages and cage rest, when needed, appear to be practical alternatives. Bicycling, weightlifting, proprioceptive training, and jumping rope, which are the mainstays of rehabilitation in people, are generally either not possible in dogs, or are performed with modifications. A practical rehabilitation alternative to enhance operative outcome is needed for dogs. Electrical stimulation provides a cost- and labor-effective means of establishing early, protected return to function, as well as

the possibility of decreased degenerative joint disease (DJD) and increased muscle mass.^{44,45}

Neuromuscular electrical stimulation is an effective means of rehabilitation of the ACL-deficient knee in people and is considered by some to be more effective than volitional exercise, especially during the early rehabilitation period. Because the CrCL-deficient dog has been used extensively as a model for ACL deficiency in humans, it logically follows that post-ACL reconstruction rehabilitation methods in humans could be applied to the dog. Rehabilitation of the stabilized knee determines the success of surgical outcome, especially in the physically active human patient. Appropriate rehabilitation seems especially important to enhancing surgical outcome in large working dogs used for hunting, obedience trials, service, tracking, or police work. Loss of function in the working dog not only affects the quality of life of the patient, but may also lead to economic losses associated with an animal's inability to work. Surgical stabilization of the stifle is the treatment of choice in large-breed dogs, but all affected stifles appear to develop DJD and joint laxity regardless of the surgical intervention.

To date only a few studies have reported the use of NMES in the rehabilitation of dogs. In a study by Johnson et al.⁴⁴ NMES was used following CrCL surgery in experimental dogs, and the portable units used in this study proved quite feasible for use in dogs. The affected thigh was clipped and scrubbed with isopropyl alcohol. A limb-positioning device was constructed to maintain the limb in a normal stance position during stimulation. The rationale for maintaining the limb in a normal stance position was that stimulating the muscles in this position would be more functional. Adhesive tape was used to fix the limb-positioning device to the lateral surface of the affected pelvic limb during the 30-minute treatment session.

Carbon electrodes (4.5×4.6 cm) were placed over the distal portion of the biceps femoris muscle, and the proximal portion of the semitendinosus muscle. Additional carbon electrodes (5.1×10.2 cm) were also positioned over the distal portion of the vastus medialis muscle and the proximal portion of the vastus lateralis muscle. Water-soluble electrode gel was applied as a coupling agent, and the electrodes were fixed in position with paper tape. Channel 1 was connected to the hamstring muscle electrodes, and channel 2 was connected to the quadriceps electrodes. The parameters for the program were as follows:

Channel 1: Ramp up—3 seconds; ON: 12 seconds;
ramp down—2 seconds; OFF: 25 seconds

Channel 2: Ramp up—2 seconds; ON: 12 seconds;
ramp down—1 second; OFF: 25 seconds

Channel 2 started 3 seconds after channel 1, automatically eliciting cyclic hamstring and quadriceps co-contraction.

A symmetric biphasic waveform of 35 pulses per second was used. Amperage was increased to the tolerance level of the animal and was reduced if gross movement was noted at the stifle joint or the animal displayed any signs of distress, including turning its head in recognition of the stimulus or becoming agitated. Amperage was reduced to a level below that which produced these signs. Each treatment session lasted 30 minutes once daily, five times per week, for 4 weeks. Outcome measures were assessed at weeks 0, 2, 5, 7, 9, 13, and 19. Outcome measures included radiographs, force plate analysis, subjective lameness scores, drawer sign, thigh circumference, ROM, palpable crepitation during ROM, and postmortem evaluation of stifle joints.

Radiographic results demonstrated that the NMES group had significantly fewer bony changes associated with osteoarthritis over time. Thigh circumference was significantly greater in the NMES group at weeks 9 and 13, and subjective lameness scores were also improved over time in the NMES group ($P < .001$). The postmortem evaluation revealed that the NMES group had less cartilage damage ($P < .07$), less palpable crepitation during ROM, and less osteophyte formation. Range of motion, force plate analysis, and joint laxity were not significantly different between the groups. Meniscal damage was significantly greater in the NMES group, however.⁴⁴

In a study by Millis et al.⁴⁵ a clinical trial was designed to compare the outcomes of postoperative physical rehabilitation to traditional postoperative care in dogs surgically treated for cranial cruciate rupture. Ten dogs receiving treatment for cranial cruciate rupture were randomly assigned to either a treatment group for postoperative physical rehabilitation, or a control group. Lateral fabellar-tibial stabilization was performed in each dog by the same surgeon. Dogs in the control group were cage rested and walked on a leash for 20 minutes twice daily for 2 weeks. Dogs in the treatment group received passive ROM exercises, NMES, and walking exercises for the same 2-week period. Thigh girth measurements and maximum angles of flexion and extension were taken every other day during the 2-week period and 4, 8, and 12 weeks after surgery. The influence of other variables, such as meniscal injury, and the degree of preexisting osteoarthritis on outcome were also assessed.

By the end of the 2-week period thigh girth was significantly larger in the rehabilitation group and remained greater throughout the rest of the study, but the difference late in the study was not statistically significant. Maximal extension of the stifle was significantly greater in the rehabilitation group beginning halfway through the initial 2-week period and remained greater throughout the rest of the study. Maximal flexion was not significantly different between the two groups. The presence of moderate to severe osteoarthritis at the time of surgery retarded recovery in both groups. Interestingly meniscal injury at the time

of surgery had no effects on any of the measured parameters in the short term. In this study, muscle mass and maximum range of stifle motion were improved with postoperative physical rehabilitation. The increased muscle mass in the dogs receiving rehabilitation may have been due, in part, to NMES.

Neuromuscular electrical stimulation has been used clinically on dogs for many purposes, including to diminish joint contractures and decrease muscle atrophy associated with a variety of disorders such as inherited Labrador myopathy, postoperative atrophy, and nerve injury. Other uses have included improving limb function, decreasing pain, decreasing muscle spasm (commonly associated with intervertebral disk disease), and decreasing edema.

Neuromuscular electrical stimulation may be valuable as part of the treatment for postoperative rehabilitation of dogs undergoing femoral head and neck excision. It may also help to improve the outcome of surgically repaired chronic hip luxations, or any other surgical procedure in which quick return of muscle mass is beneficial. Neuromuscular electrical stimulation may also be used for the conservative treatment of shoulder instability⁴⁶ or iliopsoas muscle trauma.⁴⁷

Animal Reaction and Safety

Precautions should be taken to avoid injury to the handler and animal. A muzzle should be applied and the animal placed in lateral recumbency during the initial treatment. In some cases, tranquilization may be necessary if the animal is anxious. We recommend that treatment only be given under the supervision of trained personnel.

Preparation and Electrode Placement

The hair over the area to which electrical stimulation will be applied should be clipped to lower impedance (Figure 20-9). The skin should be cleaned with alcohol before treatment. It is best to locate the motor point (the area in which the motor nerve enters the muscle) so that an adequate contraction is obtained with as low a current as possible to minimize discomfort to the patient. Usually the motor point is very predictably found in the middle portion of the muscle belly. Electrodes may be placed solely on one muscle to cause a contraction and motion at the joint the muscle acts upon (Figure 20-10), or they may be placed on opposing muscle groups to cause a co-contraction (Figure 20-11) that may simulate an isometric contraction and result in little or no joint movement, or alternating stimulation of antagonist muscles to cause contractions of the muscles and movement of the joint.

Motor point location is performed by first applying gel to the skin, and then placing the electrode over the general area of the motor point, or the point at which the nerve enters the muscle (typically in the middle of the muscle belly). The other electrode is placed further distally over the muscle. With the unit on, the proximal electrode may



Figure 20-9 Preparing the skin for treatment.

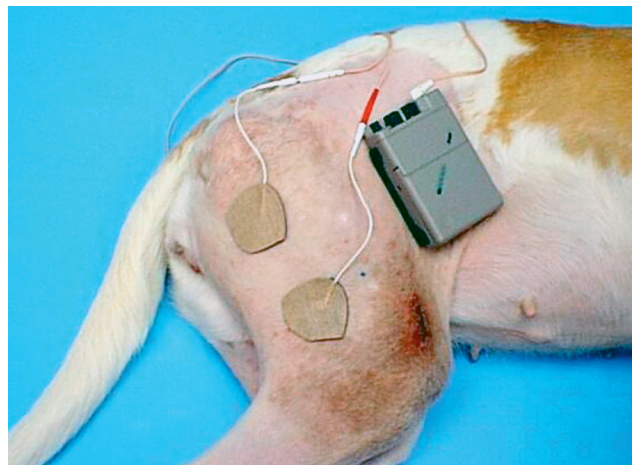


Figure 20-10 Application of electrical stimulation for contraction of the caudal thigh muscles. (Courtesy D. Millis.)

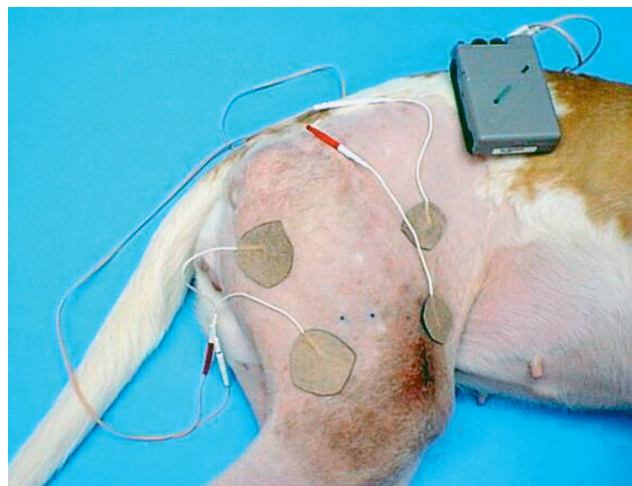


Figure 20-11 Application of electrical stimulation for cocontraction of the cranial and caudal thigh muscles. (Courtesy D. Millis.)

then be moved around until a good contraction is achieved. Setting the frequency at 1 Hz helps in motor point determination because the twitch contraction is more obvious and becomes stronger as the electrode moves closer to the motor point. An indelible marking pen may be used to draw a circle around the electrode. This allows the electrode to be placed in the same area during subsequent treatments, without having to repeat the process of motor point location. Remember to adjust the frequency to the desired setting (25-50 Hz) before beginning the actual treatment.

Treatment Time and Frequency of Treatment

Although the optimum time and frequency of treatment are unknown, most clinicians believe that electrical stimulation should be applied to the desired area(s) for 15 to 20 minutes, three to seven times per week. Occasionally a patient may experience muscle soreness early in the treatment program if electrical stimulation is used too frequently, for treatment periods that are too long, or with application of current that is too great (muscle contraction is too strong). In these cases, skipping treatment for a day or two and resuming treatment with reduced treatment time, frequency, or strength of contraction is usually adequate to resolve the problem. Co-contraction of opposing muscle groups should be considered if no joint motion is desired.

Electrical Stimulation for Pain Control

The use of electrical currents has a long tradition of pain control in humans and has been commonly used in medical practice since the 1970s. There are a variety of commonly used currents for pain control, but in all of them, surface electrodes are placed over defined locations and peripheral nerves are stimulated via the skin.

Pain-Relieving Mechanisms of Electrical Stimulation

The pain relief mechanisms of TENS are explained by multiple theories. In 1965, Melzack and Wall published their gate control theory of pain control. This theory describes how pain may be relieved by electrical stimulation. Pain signals are transmitted by small cutaneous A- δ and C fibers. If a TENS current is applied, the large cutaneous (A- β) fibers are stimulated. The signals from the A- β fibers activate inhibitory neurons in the substantia gelatinosa of the spinal cord dorsal horn and block the transmission of pain impulses from the periphery to the brain.⁴⁸ The A- β fibers can be stimulated by low current intensities and high frequencies; therefore, an electrical current at a high frequency (50 Hz or greater) is appropriate to activate this pain control mechanism.⁴⁹ Further mechanisms involve the release of endogenous opiates.⁵⁰⁻⁵² These endorphins are released from the pituitary in response to low frequency

(<10 Hz), high pulse duration (>100 μ sec) stimulation, and they produce analgesia when released. Interneurons are located in the substantia gelatinosa that produce enkephalins to inhibit the C fiber cells in the region. Collaterals of the A- δ fibers connect with these interneurons and stimulate them. If A- δ fibers are stimulated by electrical impulses, the C fiber system-type pain is decreased.⁴⁹ This is also one of the suspected mechanisms of action of acupuncture. Therefore, low-frequency, high-intensity TENS might be sufficient to stimulate the same pain control system that may occur with acupuncture. In an animal model, it was shown that high-frequency TENS increased the concentration of γ -aminobutyric acid in the spinal cord. This caused a reduction of central sensation and primary hyperalgesia.⁵³

Types of Electrical Stimulators

Many different waveforms, amplitudes, and frequencies have been recommended for pain control, and the literature describes many applications. The high-frequency, short-pulse duration (conventional) TENS (50-150 Hz, 2-50 μ sec pulse duration, low intensity) is one of the most frequently used current parameters. The intensity is increased until the dog shows some initial response, such as moving the limb, or turning the head toward the area, which we assume is a tingling sensation described by people. No muscle contraction should occur. As described the treatment is presumed to stimulate large, low-threshold A- β fibers to activate the gate control mechanism.

Acupuncture-like TENS (low-frequency, high-pulse duration, high-intensity, 1-10 Hz, 100-400 μ sec pulse duration, high-intensity) is another mode of electrical stimulation for pain control. Regarding the low frequency stimulation, some authors differentiate between the acupuncture-like TENS with frequencies between 1 and 4 Hz, and frequencies between 5 and 10 Hz. The acupuncture-like TENS is administered at an intensity that evokes visible muscle contractions. This stimulates the high-threshold A- δ fibers and C fibers that may result in the release of endogenous opioids. This form of TENS is often applied directly to acupuncture points and also to motor points in the segmentally related myotome.

Burst mode TENS is a series of pulses at 1 to 4 Hz and a high internal or carrier frequency (40 to 150 Hz), 100 to 250 μ sec pulse duration and a high intensity. The implied benefit is that it combines the benefits of both conventional and acupuncture-like TENS.

Modulated TENS offers the possibility to vary pulse length, frequency, and amplitudes automatically. This cyclic variation is believed to prevent the nervous system from becoming adapted or conditioned to the currents.

Interferential (IFC) and premodulated interferential are additional currents commonly used for pain control. These currents are very comfortable and have been commonly used in canine rehabilitation. They are delivered at a high



Figure 20-12 Application of electrodes for lumbar pain. (Courtesy D. Millis.)

carrier frequency (approximately 4000 Hz). The parameters recommended for acute pain are 80 to 150 Hz (frequency modulated), and for chronic pain are 1 to 10 Hz (frequency modulated). If the frequency modulation is not used, a fixed frequency of 50 to 100 Hz for acute pain and 2 to 4 for chronic pain is supported by the literature.⁵⁴ When using interferential current, four electrodes from two separate channels are used; the electrodes from one channel are crossed at a 90-degree angle to the electrodes from the other channel (Figure 20-12). Premodulated interferential uses just 1 channel and 2 electrodes for small areas. While it is not true interferential current, it feels almost identical and is very comfortable.

Research in Human Medicine

Transcutaneous electrical nerve stimulation has been the topic of numerous clinical trials in human medicine regarding its effectiveness for pain control in various disorders. Unfortunately the results are inconsistent because of a variety of reasons, such as weak study design, missing control/placebo groups, and small numbers of participants. The use of TENS was evaluated in several systematic metaanalyses and reviews in humans. Again the results were inconclusive, showing clear benefits of electrical stimulation for pain control in some studies,⁵⁵⁻⁵⁷ whereas others showed no effect.^{58,59} Nevertheless there is reasonable evidence in human medicine that TENS is an effective method for pain relief, especially in cases of chronic musculoskeletal pain. In one metaanalysis, the authors concluded that high frequency TENS, as well as TENS with frequencies lower than 10 Hz, produced a significant

effect and acupuncture-like TENS fell just short of significance.⁵⁴ Another metaanalysis focused on the treatment of knee OA using different types of TENS currents.⁵⁵ When the combined efficacies of conventional TENS and acupuncture-like TENS were compared with placebo, pain improved significantly in the treatment group. In addition, knee stiffness was improved in the combined treatment group. The significant pain-relieving effect was also detectable if conventional and acupuncture-like TENS were evaluated separately. Interestingly this metaanalysis found that pain relief by acupuncture-like TENS was approximately two times better than with high-frequency TENS.

Transcutaneous electrical nerve stimulation treatment is also an effective method in cases of postoperative pain. In humans who underwent lumbar canal stenosis surgery, pain and the doses of analgesics administered were evaluated in two groups. One group received pain medication and TENS treatment (80 Hz during the first 30 minutes of the therapy, switching to 5 Hz in the second half), whereas the other group received the same medication, but no TENS. In the TENS-treated group, pain and the total doses of analgesic medications were significantly reduced compared with the control group.⁶⁰

Research in Veterinary Medicine

Electrical stimulation for pain control may be used in numerous orthopedic and neurologic disorders. Studies in humans have shown benefits in acute⁶⁰ or chronic^{55,54} pain caused by osteoarthritis, spondylarthrosis, and after orthopedic and neurologic surgery. Before starting the treatment, special awareness should be given to the contraindications and precautions listed in Box 20-3. A few studies have been conducted in dogs, including the effects of TENS on osteoarthritic pain in the stifle of dogs.⁶¹ Five dogs that had chronic mild OA that was originally induced by CrCL transection and stifle stabilization were treated. Pretreatment ground reaction forces were determined using a force plate before electrical stimulation to assess the functional use of the affected limb. Immediately following these measurements, premodulated electrical stimulation (70 Hz) was applied to the involved extremity around the stifle. After treatment ground reaction forces were determined at 30-minute intervals over a 4-hour period. Each dog was also reevaluated 24 hours after treatment. Significant improvement in ground reaction forces was found 30 minutes after treatment. These differences persisted for 210 minutes after TENS application and were statistically significant 30, 60, 120, 150, and 180 minutes after treatment. The greatest improvement was found immediately after treatment. Each of the dogs was reevaluated following a 4-day rest period and still exhibited a mild increase in weight bearing on the affected extremity, but these differences were not significant. In this preliminary study, positive benefits of TENS application were apparent in dogs with osteoarthritic stifle joints.

In another study, 29 overweight dogs with lameness caused by OA of the hip, elbow, stifle, or shoulder were included in a weight loss program.⁶² The dogs were randomly divided into two groups and the diet was combined with physiotherapy. Owners of dogs in both groups were instructed to perform massage, passive range of motion, and gradually increase levels of controlled exercise. Group 1 dogs were additionally treated with TENS in a physiotherapy clinic twice weekly. Body weight, subjective lameness, and pain scores were evaluated monthly. A force-plate analysis was performed every second month. Asymmetric weight distribution between limbs affected and unaffected by OA was demonstrated using symmetry indices, calculated using peak vertical force (PFz) and vertical impulse (IFz). During the first 2 months of the study the animals in both groups lost about the same amount of body weight, but at day 90 the TENS-treated group had lost significantly more weight than group 2. This difference was also noted at the 6-month point. During the course of the study the lameness scores decreased significantly, beginning at day 30 in group 1 and at day 60 for group 2. Pain scores also decreased, with the difference reaching significance sooner in group 1 (at 60 vs. 90 days). Force plate analysis showed that symmetry indices for PFz and IFz in group 1 were significantly improved at each evaluation. In group 2, only the symmetry index for PFz improved, and only at day 120. At day 60, there was no difference in body weight between groups, but the symmetry indices were only significantly improved in group 1. This suggests that TENS could have a significant influence on lameness. The authors concluded that a combination of dietary management and intensive physiotherapy including TENS treatments may effectively reduce disability in dogs that are overweight and have osteoarthritis. Further study of TENS application in dogs, including different parameters and altered physical activity, may reveal additional benefits.

Preparation and Electrode Placement

The hair (unless very short) should be clipped before the treatment if rubber electrodes or gel pads are used. If electrodes are not coated with gel, suitable contact medium, such as ultrasound or electrocardiogram gel, should be applied in an even layer to ensure uniform transmission of the current to the skin. Clipping should be performed carefully to avoid skin damage. Another possibility is the use of pads with brushlike needle pads. In this case, clipping the hair is not necessary, but the skin should be wetted with water or alcohol. For each treatment session the patient should be placed in a relaxed position (e.g., lateral recumbence) on a soft and comfortable surface.⁶³ To achieve a beneficial effect, the electrodes should be placed in the proper location. Several possible locations may be chosen, and the choice should be adapted based on the effectiveness of the treatment. Sometimes it is necessary to try

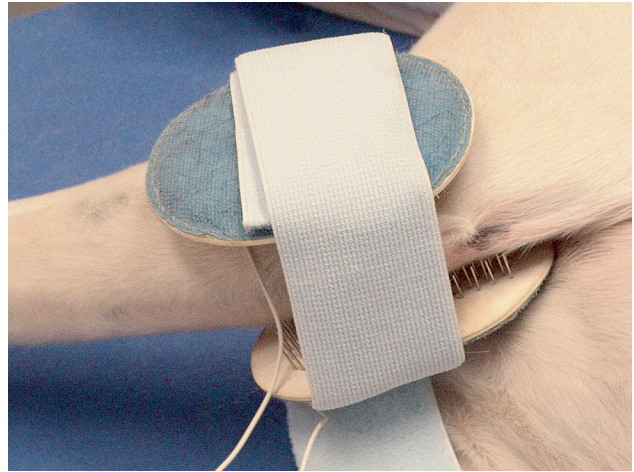


Figure 20-13 Application electrodes for chronic pain using flexible E-pads because of elbow dysplasia. (Courtesy B. Bockstaler.)



Figure 20-14 Application of electrodes for lumbar pain utilizing flexible E-pads. (Courtesy B. Bockstaler.)

several locations to achieve the best pain-relieving effect. In a study of human long-term users, most patients applied the electrodes over or directly proximal to the painful area.^{64,65} In a dog one example would be placement over the medial and lateral aspects of an arthritic elbow (Figure 20-13) or stifle joint, or along the edges of the painful area (e.g., cranial and caudal region of painful back muscles) (Figure 20-14). This so-called “local placement” is mainly used in chronic conditions.

In cases of acute disease (e.g., an acute stage of OA) if placing the electrodes over the area is painful, or aggravates the symptoms, segmental stimulation (stimulation within a dermatome) may be applied with the electrodes placed to the left and right of the vertebral column near the nerve origins of the target region.⁶³ To place the electrodes correctly an in-depth knowledge of canine neuroanatomy is necessary. For example, to treat hip joint pain the pads could be placed over the L4-S1 nerve roots on the side of

the painful hip. Other possibilities include placement of electrodes over acupuncture or trigger points, or along the peripheral superficial nerve tracts.

Intensity, Treatment Time, and Frequency of Treatment

The intensity must always be adapted to the needs of the individual patient and their response to the treatment. To evaluate treatment success the pain level of the animal should be evaluated initially and during the course of treatment. This is necessary to monitor the effectiveness of the selected parameters, such as electrode position, intensity, and frequency, and the recovery of the animal. The intensity can be chosen according to the subjective evaluation of the patient, but it also depends on the desired pain-relieving mechanism and the type of TENS.

To make the animal comfortable with TENS the treatment is often started with an intensity of the current just below the sensory response threshold. In dogs this can be achieved by carefully increasing the intensity until the animal shows a reaction, such as glancing at the electrodes or tensing the limb muscles. The intensity is then decreased slightly just to the point that the animal stops reacting. Especially when treating animals it is always mandatory to start the treatment carefully and with patience. Each patient should have time to adapt to the treatment and relax. With increasing intensity the patient likely will feel a tingling sensation. This should be neither painful nor should it cause muscle contractions. A higher intensity is often not recommended and not tolerated by the animal. Accordingly a distinction is drawn between intensities below the motor threshold (no visible muscle contractions), at motor threshold (visible twitch contraction), and above it (wavelike muscle movements and contractions).

There are different recommendations for treatment duration. For example, for treatment of OA using conventional TENS, treatment times vary between 15-,⁶² 30-,^{61,66,67} and up to 60-minute⁶⁸ sessions. Also, the timing of the treatment varies from three sessions daily for 3 weeks,⁶⁸ to twice daily for 6 weeks, to twice weekly for 4 weeks.⁶⁹

In general, the following recommendations may be applied.⁶³ In acute stages, a lower-intensity, short treatment duration, and short intervals are suitable. Therefore, in an acute case, it is possible to treat the animal one to two times daily, with a 15-minute treatment duration. In chronic conditions, higher intensities with longer treatment durations and intervals between the treatments are recommended; for example, treatment two to three times per week, for 5 to 6 weeks, with each session lasting 30 minutes.

Future Areas for Study

Continued clinical trials investigating electrical stimulation protocols for muscle strengthening and pain control need to be conducted. Given the similarities in neurologic and

orthopedic conditions between dogs and people, it seems that the indications for electrical stimulation are similar. Iontophoresis is another area that should be studied in dogs to investigate its possible clinical application.

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21 Laser Therapy in Canine Rehabilitation

Darryl L. Millis and Debbie Gross Saunders

Healing properties have been attributed to light for thousands of years. Light provides electromagnetic radiation in the form of photons. A number of forms of light have been used for therapeutic purposes, including sunlight, incandescent light, infrared light, ultraviolet light, light-emitting diodes (LEDs), and, recently, therapeutic lasers. This chapter focuses on therapeutic lasers rather than surgical lasers. Laser therapy has been increasingly incorporated into rehabilitation programs for a variety of conditions, including skin wounds; muscle, tendon, and ligament injuries; neurologic conditions; arthritis; and pain. Many studies have indicated encouraging results with laser therapy. The principles of lasers, research pertaining to the use of laser therapies in rehabilitation, and the basics of laser therapy are addressed.

Laser Therapy

The term *laser* is an acronym for *light amplification by stimulated emission of radiation*. The concept of the use of light for therapeutic purposes, called *phototherapy*, originated from the belief that sun and other sources of light, such as infrared and ultraviolet light, have therapeutic benefit. Many different types of lasers are available for medical and industrial purposes. Low-power laser devices, a form of artificial light, were first used as a form of therapy more than 30 years ago. Today, a variety of lasers are in use for various purposes.

The initial types of lasers used for rehabilitation purposes, commonly known as low-level laser therapy (LLLT), are also called *cold lasers*. In contrast, surgical lasers are high power and capable of thermal destruction of cells and tissues. Recently another form of laser, known as a *therapeutic laser*, has been introduced for rehabilitation purposes and delivers more power than low-level lasers, but less power than surgical lasers. Therapeutic lasers have become increasingly popular in both small and large animal rehabilitation for a variety of conditions.

The lasers used in rehabilitation help to modulate cellular functions. This process is known as *photobiostimulation* and is defined as nonthermal interaction of monochromatic radiation with a target site.¹ Although the physiologic interaction of this type of energy application

on tissue is still not completely understood, low-energy lasers have been reported to modulate various biologic processes, such as mitochondrial respiration and adenosine triphosphate (ATP) synthesis, to accelerate wound and joint healing, and to promote muscle regeneration.^{2,3} In addition, acute and chronic pain control has been reported using this type of low-energy photon therapy.⁴ Treatment of chronic and acute edema, neurologic conditions, and postoperative care are some other popular conditions treated with laser therapy.

Properties of Lasers

Basic light sources emit electromagnetic radiation that is visible to the normal eye. Although natural light sources, such as sunlight, are forms of electromagnetic radiation, lasers are artificial sources that emit radiation in the form of a flow of photons (Figure 21-1). The process of light emission begins with activation of electrons in the laser unit, generally either helium-neon (HeNe), gallium-arsenide (GaAs), or gallium-aluminum-arsenide (GaAlAs) to an excited state.⁵ When the electrons drop from their excited state to their ground state, photons are emitted. Although some photons are absorbed by the laser chamber wall, others stimulate the emission of other photons, and together they travel in the chamber, amplifying this stimulated emission, which results in a chain reaction. Some of these photons are released through a semireflective mirror to form a beam of light.

The major difference between laser light and light generated by normal sources is that laser light is monochromatic, coherent, and collimated. *Monochromatic* means that all light produced by the laser is of one wavelength, and therefore a single color. Sunlight, or white light, may be broken into several different colors of different wavelengths by a prism (Figure 21-2). Laser light has electromagnetic radiation of only one wavelength. Commercial lasers occasionally have two or more different wavelengths within a single unit to achieve different effects, but each component within the unit has a single wavelength and is monochromatic. The coherent properties of light mean that the photons travel in the same phase and direction (Figure 21-3). Laser light is also collimated, which means that

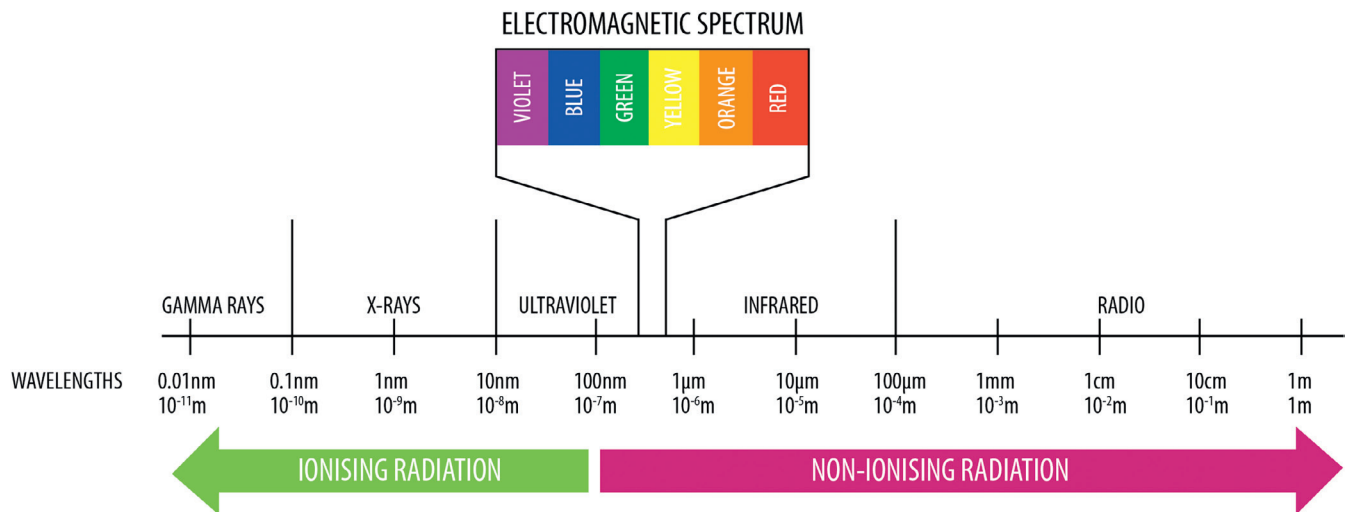


Figure 21-1 Electromagnetic spectrum. Laser light typically falls within the infrared or near red portion of the electromagnetic spectrum. (Photo Courtesy Companion Laser, LiteCure LLC, Newark, Delaware.)

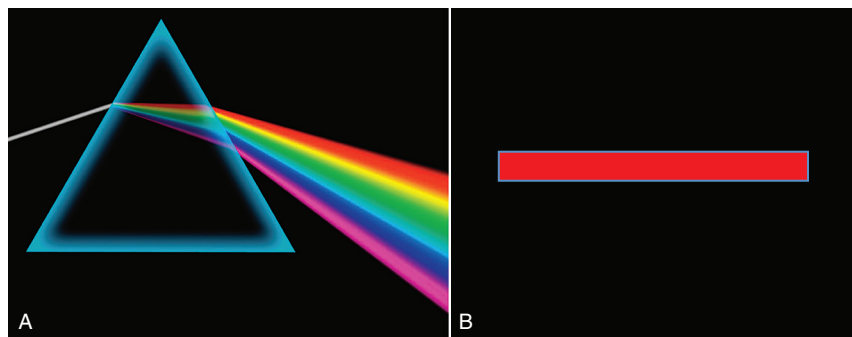


Figure 21-2 A, Sunlight, or white light, may be broken into several component colors with different wavelengths. B, Laser light is monochromatic and is produced by a single wavelength.

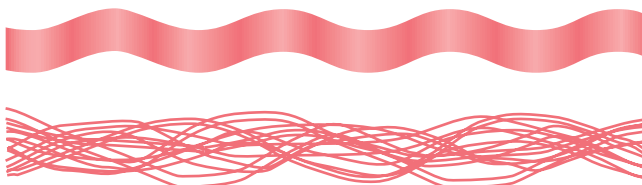


Figure 21-3 Laser light is coherent, with the photons traveling in the same phase and direction, whereas normal light is incoherent.

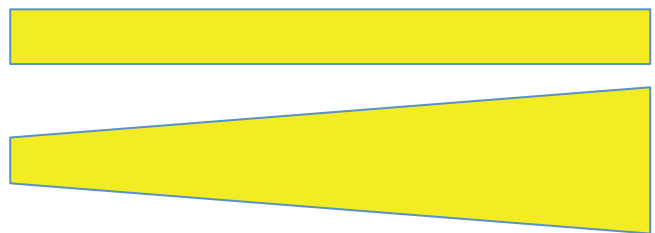


Figure 21-4 Laser light is collimated, which means that there is minimal divergence in the laser beam over a distance. Normal light is not collimated, and the beam diverges over a distance.

there is minimal divergence in the laser beam over a distance (Figure 21-4). Using a monochromatic light source allows the absorption of the light to be targeted to specific wavelength-dependent chromophores, or photon acceptors.⁵ The properties of coherence and collimation allow the light to be focused precisely on small areas of the body. These properties allow low-level laser light to penetrate the

surface of the skin with no heating effect, no damage to the skin, and few or no side effects when properly used.

Laser light interacts with tissues in various ways. Light may be reflected, scattered, transmitted, or absorbed. Reflected photons have no clinical effect and may be dangerous to tissues that encounter reflected photons, such as the eyes. In addition to surfaces such as watches, tables,

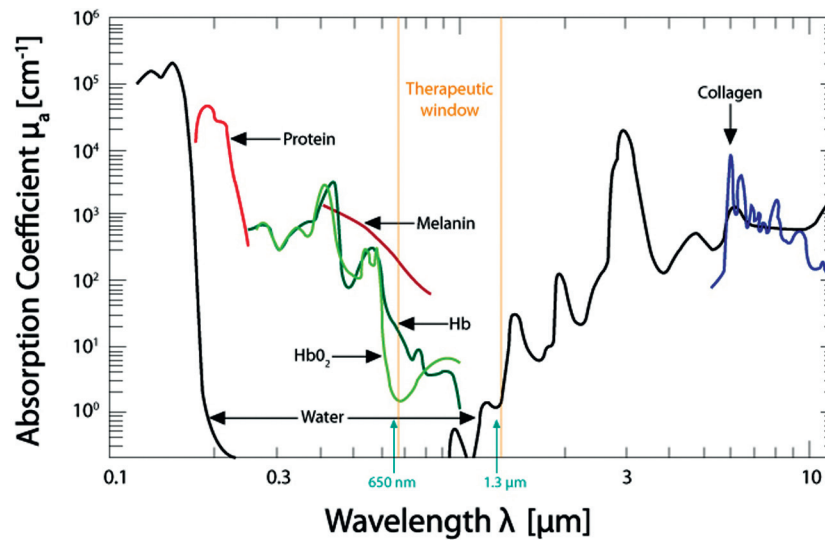


Figure 21-5 The wavelength of the photons is important in laser therapy. Various substances preferentially absorb laser light at different wavelengths. (Photo Courtesy Companion Laser, LiteCure LLC, Newark, Delaware.)

and instruments, the epidermis is responsible for reflecting most of the photons from the skin. To reduce reflection of photons, the laser beam should be directed as nearly to 90 degrees as possible to the skin surface. As photons pass through the tissues, some are scattered. Each photon that is scattered when striking an object reduces the amount of energy that can be directed at the target tissue. Scattering decreases as the wavelength increases because longer wavelengths penetrate deeper into tissues. Transmitted photons pass completely through the tissue without being absorbed. This is rarely a problem in rehabilitation because the tissues are generally thick enough to prevent complete transmission. Finally, photons may be absorbed. Photons are absorbed by chromophores (molecules that absorb certain wavelengths of light) and it is in this manner that laser may affect tissues. The most common chromophores are water, hemoglobin, melanin, cytochrome C system in mitochondria, proteins, and amino acids.⁶ The relative amounts of chromophores vary among tissues, and the absorption of photons by chromophores varies, in part, on the wavelength.

The wavelength of the photons is important in laser therapy. Wavelengths are measured in nanometers (nm) and determine, in part, the biologic effect on tissues. For example, ultraviolet light (100-400 nm) is absorbed primarily by melanin, proteins, and nucleic acid; visible light (400-760 nm) is scattered and absorbed, with the absorption primarily by melanin, hemoglobin, and myoglobin; with near infrared light (760-1400 nm), photons are mainly scattered, but a variety of chromophores absorb these photons, although somewhat weakly; in the far infrared zone (1400-10,000 nm), absorption is almost entirely by water (Figure 21-5).⁶ Therefore the optimal window to

have tissue penetration with less scattering and surface absorption is likely to be in the 600-1200 nm range.

In addition to wavelength, the power of the laser is also a characteristic that is important. Power is a unit of time and is expressed in watts (W) or milliwatts (mW).

$$1 \text{ watt} = 1 \text{ joule/second}$$

The spot size of the unit indicates the surface area the laser covers while held in a stationary position. This is also known as the *power density*, or *intensity*, and indicates the power per surface area unit. It is usually indicated by watts/cm². Larger spot areas result in a more homogeneous passage of laser light through tissues with less photon scattering and light dispersion.

The energy is the power emitted over time, and is usually measured in joules. This is also frequently used to report the dosage of laser light.

$$1 \text{ joule} = 1 \text{ watt} \times 1 \text{ second}$$

For example, a 50-mW laser delivers 1 J of energy in 20 seconds of treatment time. A 500 mW laser takes 2 seconds of treatment time to deliver 1 J of energy. A 1-W laser takes only 1 second, and a 10-W laser takes $\frac{1}{10}$ of a second. From a treatment standpoint, a higher watt laser delivers the treatment in a shorter time. If an arthritic stifle is being treated, it may require a total dose of 100 to 500 J depending on the size of the dog and treatment area. So the treatment time may vary greatly depending on the unit used.

The energy density, or radiant exposure, is the energy per surface unit and is typically indicated by joules per cm², or J/cm². This term is also clinically called the *dose of laser energy* and is important in determining and describing treatment protocols. To date, only the dose of laser energy

Box 21-1 Biologic Effects of Laser Therapy

Activation of respiratory chain enzymes, especially cytochrome C oxidase
 Oxygen production
 Formation of proton gradients across cell and mitochondrial membranes
 Adenosine triphosphate production
 DNA production
 Cell proliferation
 Reduced cyclooxygenase and prostaglandin E₂ production

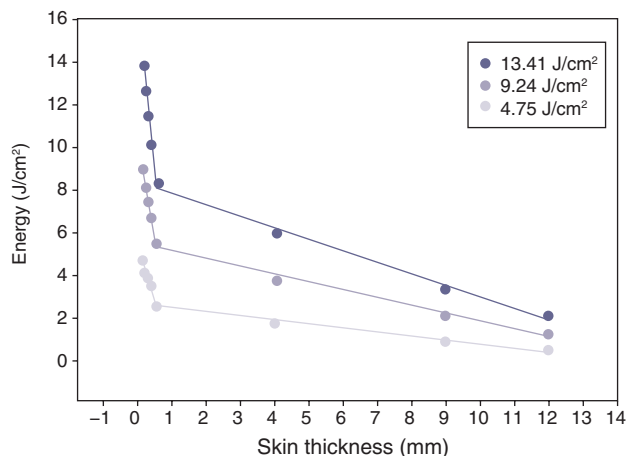


Figure 21-6 Three different doses of laser light and penetration through skin. Greater doses of laser energy penetrate deeper into tissues. (From Topping A, Gault D, Grobbelaar A et al: Does Low Penetration of Human Skin by the Normal Mode Ruby Laser Account for Poor Permanent Depilatory Success Rates? *Lasers Med Sci* 16:224-229, 2001.)

has been shown to have biologic effects as compared with the rate of administering laser energy, or the power of the laser (Box 21-1). Greater doses of laser energy penetrate deeper into tissues (Figure 21-6). The Lambert-Beer law indicates that for homogenous tissues with a constant coefficient of absorption, more photons will reach a given tissue depth with a greater dosage, and the intensity of light passing through tissues decreases exponentially with increasing tissue depth. There is a great deal of difference among lasers related to the power, with those delivering energy at a greater rate generally costing more. Although there is not thought to be an advantage in efficacy regarding low- versus high-power lasers, the dosage (total joules) can be administered in a much shorter time with a high-power laser, therefore reducing labor costs. In addition, high-power lasers are generally administered in a sweeping fashion, whereas lower power lasers are usually held in place until the administration of the dose for a particular spot treatment is complete. The sweeping motion may result in more thorough coverage of the treatment area as compared with the spot treatment of lower power lasers.

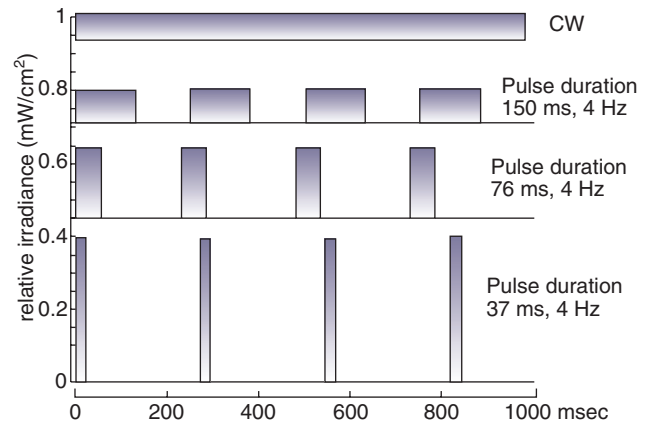


Figure 21-7 Diagram of laser light delivered as continuous wave or pulsed laser light of various pulse durations. (Based on a diagram from Hashmi JT et al: Effect of pulsing in low-level light therapy. *Lasers Surg Med* 42:450-466, 2010.)

The sweeping motion also allows the clinician to cover other surrounding areas that may be causing secondary or tertiary pain. For example, when treating canine hip dysplasia to address pain and inflammation, the coxofemoral joint should be treated from all areas—medial, lateral, caudal, and cranial. With the sweeping motions covering the four areas, the laser may affect the surrounding soft tissues that are often painful in clinical canine hip dysplasia.

Lasers can emit photons continuously or in a pulsed fashion (Figure 21-7). With continuous wave laser, radiation is emitted at a constant power over the entire treatment time. With pulsed wave laser treatment, impulses may be emitted at varied rates, with an on time when energy is emitted (pulse width or pulse duration [PD]) and an off time when there is no energy emitted. The duty cycle (DC) is the percentage of time that radiation is emitted in relation to the total on-off time. The pulse rate, or frequency (F), is measured in Hertz (Hz). The relationships between these measures may be expressed as:

$$DC = F \times PD$$

One report reviewed the literature regarding pulsed versus continuous laser application.⁷ Although the authors concluded that there is some evidence that pulsed laser has different effects than continuous laser, further work is necessary to define optimal treatments for various conditions, and to determine the optimal pulse structures. One theory behind the use of pulsed laser is that with relatively short pulses, the laser could be excited to higher levels as compared with continuous wave laser, in which thermal damage to the tissues might limit the maximum amount of energy that can be applied. With short pulses, the thermal effects on tissues might be reduced, limiting tissue damage. Pulsed GaAs and indium-gallium-arsenide (InGaAs) lasers may allow for deeper tissue penetration without thermal damage, as well as allowing for shorter treatment times. In addition to decreasing the thermal effects on tissues, pulsing of the

laser may also have some resonance with certain functions, such as brain waves and opening of cellular ion channels. Pulsing may also have effects on photodissociation of substances, such as nitric oxide (NO) from protein binding sites, allowing multiple episodes of dissociation. Although the actual depth of penetration is probably not that different in pulsed versus continuous wave lasers, for many conditions, depth of penetration is a less important issue, such as with treatment of elbow or stifle arthritis. One author concluded that pulsed lasers (904 nm) were not significantly more effective than continuous-wave lasers (810-830 nm) for treatment of tennis elbow in people.⁸ Both types were equally effective, but half the energy was needed with pulsed lasers. However, a review of literature comparing continuous wave laser with pulsed laser identified six of nine studies that found pulsed wave to be more effective.⁷ A clinical study of wound healing and experimental studies of pain management and stroke recovery have suggested that pulsed laser has better results than continuous laser. Another study comparing continuous and pulsed laser found both to be equally effective. Only two of the nine studies reported better results with continuous than pulsed laser, although both treatment methods achieved better results than placebo treatments. Some limitations of the studies in this review were that the same parameters of laser wavelength were not used in some instances, making direct comparisons difficult. One study found a combination of pulsed and continuous laser to be more effective than either alone for stimulating nerve regeneration in a median nerve transection and repair study,⁹ whereas most studies have shown continuous laser to be more effective than pulsed laser for nerve conduction and regeneration.⁷ Studies comparing various rates of pulsed laser treatment show no consistent results, with some demonstrating better results with lower frequencies and some with higher frequencies.⁷

Classification of Lasers

The American National Standards Institute in the United States and the International Electrotechnical Commission have defined classes of lasers based on their ability to damage tissues. In general it is the thermal damage that injures tissues, especially the eyes, and in some cases, skin. Even small amounts of laser light can cause permanent damage to the retina.¹⁰

Current classifications of lasers are related to the wavelength and the maximum output power (in watts) or energy (in joules). These classifications allow the placement of various lasers into categories based on their ability to cause tissue damage, especially to the eyes. For example, Class 1 lasers are very mild and are safe under all conditions of normal use, and include supermarket scanners and post office readers. Class 1M lasers are safe for all uses except when passed through a lens, such as a microscope or telescope. These types of lasers have large-diameter beams or



Figure 21-8 Class IV laser.

divergent beams and cannot achieve the power to cause damage unless the laser is focused with a lens. The power is less than that of a Class 3B laser, but cannot damage the retina unless the light is focused. Class 2 lasers are in the visible-light spectrum (400-700 nm) and include items such as laser pointers, and occasionally therapy lasers. Because these lasers are visible light, the blink reflex generally limits exposure to less than 0.25 seconds. Class 2 lasers are limited to 1 mW continuous wave, or more if the emission time is less than 0.25 seconds or if the light is not spatially coherent. Class 2M lasers, similar to Class 1M lasers, have a large-diameter light beam or the beam is divergent and is safe unless the light is viewed through an optical instrument.

Class 3B lasers are either continuous light in the 315 nm to far infrared ranges limited to 500 mW, or pulsed lasers 400-700 nm wavelength limited to 30 mW. When using Class 3B lasers, protective eyewear is required if direct viewing of the light may occur. Class 3R lasers are considered to be safe with restricted exposure and are visible-light continuous lasers with power limited to 5 mW.

Class 4 lasers have the greatest potential to cause tissue damage and include all lasers with power greater than that of Class 3B lasers (Figure 21-8). Surgical, industrial cutting lasers, and some therapeutic lasers are Class 4 lasers. Surgical lasers typically have power between 30 to 100 W, whereas therapeutic lasers may be 1 to 15 W. Class 4 lasers may burn the skin or cause permanent eye damage as a result of direct, diffuse, or indirect beam viewing, such as might occur with reflection of the beam, even from matte surfaces. Therefore, the therapist must use great care to control the beam path and protective eyewear must be worn by all in the immediate area. The U.S. Food and Drug

Administration (FDA) requires all Class 3B and Class 4 lasers in the United States to have safety features, including a key switch, a safety interlock dongle, a power indicator, an aperture shutter, and an emission delay (normally 2-3 seconds). In addition, the U.S. Occupational Safety and Health Administration requires the use of adequate eye protection when eye exposure may occur while operating lasers in these classes.

Lasers Used in Physical Rehabilitation

Most lasers used in physical rehabilitation are Class 3 or 4 lasers and have a finite lifespan that generally varies from 5000 to 20,000 hours. LLLT devices are typically Class 3 lasers and are low power, often less than 100 mW, and they do not heat tissues. The lack of heat generation is significant. If heat is generated by a Class 3 laser, the unit may not be functioning properly or the device may be improperly used. LLLT devices typically have small treatment beam diameters, up to 1 cm. The basic types of lasers used for LLLT are gaseous HeNe and GaAs or GaAlAs semiconductor or diode lasers. HeNe lasers have a visible red light with a wavelength of 632.8 nm, whereas GaAs and GaAlAs have invisible light near the infrared band with a wavelength of 820-904 nm.

The wavelength of the laser is important because it determines, in part, the laser's effect. Longer wavelengths are more resistant to scattering than shorter ones. Therefore, GaAs and GaAlAs lasers penetrate more effectively (direct effect up to 2 cm, indirect effect up to 5 cm) than HeNe lasers (direct effect up to 0.5 cm, indirect effect up to 1 cm) because there is less absorption or scattering in the epidermis and dermis. Light waves in the near infrared ranges penetrate the deepest of all light waves in the visible spectrum (Table 21-1). Although this spectrum of light is not visible, commercial lasers should have an LED that allows the therapist to see where the laser light is aimed.

Lasers can only have an effect if they stimulate cells, and the depth of laser penetration is important to determine if cells at a particular depth may be stimulated. Very few studies have measured the depth of laser penetration through skin, and these studies have been done primarily

on human skin samples. Readers should use caution in interpreting the results of these studies and extrapolating them to dogs because of inherent differences in skin thickness, skin composition, cutaneous blood supply, skin color, and thicker hair coats. Lasers with shorter wavelengths are most effective for surface-level conditions because they do not penetrate to deep tissues or large joint capsules. Wound care and wound healing should benefit from lasers with shorter wavelengths. Light that is not absorbed by water, hemoglobin, or melanin is gradually attenuated as it passes through tissues. The level of scattering and absorption is such that HeNe (632.8 nm) laser light loses approximately one third of its intensity during the first 0.5-1 mm of tissue depth.¹¹ The depth (in centimeters) at which the energy of a laser beam is 36% of its original values is termed the *first depth of penetration*.¹² This attenuation of energy is derived by dividing the original value by a constant, 2.78. Subsequent depths of penetration may be determined by dividing by 2.78 again, so that the level of energy at the second depth of penetration is 13%. However, because biologic effects may be noted with relatively low energy (0.01 J/cm²), lasers that typically deliver 1 to 4 J/cm² may penetrate up to 0.5 to 2 cm before the energy level is so low that they have no effect. These factors apply to human skin, and, although there are many similarities between human and animal skin, there are significant differences caused by hair, pigmentation, and tissues.

One study evaluated the penetration of ruby laser light (694 nm) in human white skin. Doses of 4.75 J/cm², 9.24 J/cm², and 13.41 J/cm² were administered to the skin.¹³ (see Figure 21-6). The most likely explanation for the tremendous drop in energy in the first mm is scattering of laser light, which supports the hypothesis that the greatest hindrance to a photon energy beam traversing the skin is from the stratum spinosum of the epidermis, collagen in the dermis, or both. Beyond the depth of the dermis and at the levels of the subcutaneous fat, the energy drop per distance traveled was not as great. The mean maximum depth of penetration was 14.8 mm which appeared to be a function of wavelength and not dosage. The natural chromophore for this wavelength of laser is melanin. Therefore the depth of penetration might be expected to be less in dark-skinned dogs. After passing through the skin, penetration through the muscle becomes easier because muscle has a smaller scattering coefficient.¹⁴ In fact, transmission of laser light between 600 and 800 nm is approximately fourfold greater in skeletal muscle than skin.¹⁵

Another study with laser of similar wavelengths had similar results. An HeNe laser (632.8 nm, 50 mW power) and a semiconductor laser (675 nm, 21 mW power) were used to measure transmission in human skin and skin with granulation tissue from leg wounds with ulcers.¹⁶ In the thickest sample (epidermis, dermis, and subcutaneous tissue, 2 cm thickness), approximately 0.3% of HeNe laser and 2.1% of semiconductor laser light penetrated through

Table 21-1 Wavelengths of Various Components of the Electromagnetic Spectrum

AM Radio	10,000 cm
TV and FM	100 cm
Microwave	10 cm
Infrared	700 nm
Ultraviolet	10 nm
X-rays	1 nm

all layers. Penetration of laser light was greatly attenuated at 1 to 3 mm of tissue depth. Transmission in granulation tissue was approximately 2.5 times higher than that in normal skin.

In another study of depth of penetration, a GaAlAs laser (wavelength 850 nm, 100 mW power) was applied to human abdominal skin samples.¹⁷ The intensity of laser radiation was reduced by 66% after being transmitted through a 0.784 mm sample of human abdominal tissue. In this study, most laser radiation was absorbed within the first 1 mm of skin. Additional research is needed in small animals to determine the depth of penetration with different hair coats and skin color.

Biologic Effects of Laser Therapy

Most studies of laser use in rehabilitation have centered on wound healing and pain management. However, information regarding their efficacy in reducing pain or promoting tissue repair is incomplete.¹⁸ Recently, interest has been generated in the United States regarding their use in treating people, and a natural extension has been an interest in treating animals. In evaluating the potential usefulness of laser therapy in rehabilitation, the reader is encouraged to be critical of studies that have been performed, and to evaluate these studies in light of recent advances in laser technology and the application of new information. Until recently, low-energy and therapeutic lasers were not approved for medical treatment in this country. However, as more evidence becomes available, they will likely be increasingly used.

Most of the potential responses of cells and tissues to laser energy have been studied in *in vitro* models. Photons delivered to the cells and tissues trigger biologic changes within the body. Photons are absorbed by chromophores and respiratory chain enzymes (especially cytochrome C oxidase, the terminal enzyme of the mitochondrial respiratory chain) within the mitochondria and at the cell membrane. Copper components of cytochrome C oxidase are photoacceptors.¹⁹ Cellular signaling causes a cascade of cellular reactions resulting in things such as NO dissociation from cytochrome C oxidase.¹⁹ This, in turn, results in further changes down the respiratory chain. Oxygen production and the formation of proton gradients across the cell and mitochondrial membranes may also occur. The enzyme flavomononucleotide is activated and initiates the production of ATP. Even small changes in ATP levels can change cell metabolism. The role of ATP in cellular energetics is well known, but ATP may also act as a signaling molecule to enhance cell to cell communication.¹⁹ ATP may bind with the cell receptor P2X, which opens a channel to allow sodium and calcium to enter cells, resulting in a cascade of intracellular interactions. Increased intracellular calcium positively affects mitochondrial function. ATP may act as a neurotransmitter when released by nerve cells, but may also have other effects when released by other cells, such as bone production and cell proliferation.²⁰ The

changes in ATP may, through its effects as a neurotransmitter, explain some of the effects of laser therapy on pain modulation and the effect of acupuncture.

The effect of laser treatment on various cells *in vitro* has been evaluated in several studies.²¹ HeNe laser (632 nm) with a power of 10 mW and a dose of 0.43 J/cm² resulted in greater proliferation and differentiation of human osteoblasts *in vitro*.²²

DNA production may also be stimulated. Photons also appear to affect tissues by activating enzymes, which trigger biochemical reactions in the body. Because cellular metabolism and growth are stimulated, lasers have the potential to accelerate tissue repair and cell growth of structures, such as tendons, ligaments, and muscles. Although lower dosages of laser energy appear to stimulate tissues, higher dosages may actually inhibit responses such as tissue healing. The enzyme kinetics may differ between high enzyme/substrate, and low enzyme/substrate conditions.²³

A great deal remains unknown regarding the mechanisms of laser light in biologic systems. For example, NO and carbon monoxide inhibit cytochrome C oxidase. The effects of near infrared laser radiation in conjunction with NO on cytochrome C oxidase are unknown.²⁴ Further work is also needed regarding the effects of temperature, pH, exposure times, and frequency of photobiomodulation treatment on cytochrome C oxidase.

Laser therapy may also stimulate stem cell proliferation.¹⁹ In addition to the direct effects of laser at a particular depth of tissue, indirect effects may also be seen. These cellular and tissue effects are decreased in the deeper tissues, and are catalyzed by the energy absorption in the more superficial tissues.

Laser therapy may also have antiinflammatory effects, similar to nonsteroidal antiinflammatory drugs (NSAIDs) and steroids. Studies of laser therapy in cell culture have indicated that inflammation can be reduced by a decrease in prostaglandin E₂ (PGE₂) and cyclooxygenase-2 (COX-2) concentrations.^{25,26} This effect has also been demonstrated in an animal model,²⁷ as well as people with Achilles tendinitis.²⁸ Frequently, owners seek alternatives to NSAIDs for their animals, and laser therapy may offer alternative care. Animals that are unable to consume NSAIDs may also benefit from laser therapy as an alternative for inflammation.

Research Regarding Laser Therapy

Wound Healing

Laser light stimulates fibroblast development and may affect collagen production to repair tissues. Laser light may also accelerate angiogenesis and increase the formation of new capillaries in damaged tissues, possibly improving the rate of wound healing (Box 21-2). Therefore laser therapy may aid healing of open wounds and burns. There is an increased growth factor response within cells and tissues, which may

Box 21-2 Laser Therapy Research**Wound Healing**

- Fibroblast stimulation
- Capillary formation and angiogenesis
- Collagen formation
- Enhanced adenosine triphosphate, protein, and growth factor production
- Vasodilation
- Lymphatic drainage
- Potential inhibition of wound healing at high doses

Bone and Cartilage

- Enhanced early bone repair
- Increased collagen deposition and bone trabeculae
- Adjunct to treatment of osteomyelitis
- Fibrous healing of cartilaginous defects
- Improved maintenance of cartilage in immobilized joints

Arthritis

- Inhibition of inflammation
- Inhibition of cyclooxygenase-2 enzyme and prostaglandins
- Reduced pain
- Possible reduction of morning stiffness in rheumatoid arthritis

Ligament and Tendon Conditions

- Pain reduction in acute tendonitis
- Improvement with lateral epicondylitis
- Reduced pain and inflammation in Achilles tendonitis
- Improved collagen organization
- Improved biomechanical properties

Analgesia

- Reduced pain in postoperative incisions

Inhibits Nociceptors

- Potential reduction in transmission of pain signals to pain centers in the brain
- Increased release of endorphins and enkephalins
- Stimulation of trigger points and acupuncture points
- Slowing of nerve conduction velocity
- Reduced action potential
- Suppressed substance P
- Disruption of axonal flow

Peripheral Nerves and Spinal Cord

- Promotion of nerve recovery after injury
- Increased axonal sprouting and growth
- Increased myelination
- Reduced degeneration of neurons
- Increased growth-associated protein-43 and calcitonin gene-related peptide

be related to increased ATP and protein synthesis. Laser light therapy causes vasodilation and also may improve lymphatic drainage. This may result in decreased edema and swelling caused by bruising or inflammation.

In cell culture, doses of 0.5 to 16 J/cm² were evaluated in vitro as a single exposure on two consecutive days.²⁹ A single dose of 5 J/cm² resulted in increased proliferation

and viability of injured fibroblasts. Higher doses (10 and 16 J/cm²) had reduced cell proliferation and viability, with damage to the cell membrane and DNA. The results of this in vitro study do not necessarily imply that similar findings would be present in vivo, but do suggest that there may not be a linear dose-response effect.

A review of experimental studies related to laser treatment of wounds in 8 mouse and 39 rat model studies has been reported.³⁰ Wound healing in rodent models differs from wound healing in people because of the loosely attached skin in rodents as compared with people, making wound contraction a significant factor in healing in rodents. The loosely attached skin of rodents may be similar to dogs and cats, and therefore mice and rats may be a useful model of wound healing to evaluate treatments for dogs and cats. Many studies continue to inadequately report laser parameters used in treatment protocols. In spite of these shortcomings, the results of these studies suggest that laser or monochromatic light is effective as a treatment for wounds when applied at appropriate doses. Coherence of light may not be an important factor when treating surface wounds. LEDs with monochromatic light appear to have effects on topical wounds, similar to those obtained with laser light. Wavelengths in both the visible red (630-685 nm) and infrared (700-1000 nm) typically stimulated wound healing in a variety of models. Some studies suggested that the best results were obtained when using 660-nm laser light in the early stages of wound healing, whereas 780 nm produced positive results throughout all stages of wound healing.^{31,32} In general, there was no clear association between outcomes and power levels used in the studies evaluated in this review.³⁰ However, lack of standard experimental design, including total dose administered, made it difficult to draw conclusions. In those studies that reported energy density or when calculation was possible, there appeared to be a dose-response relationship.³⁰ However, there may be a biphasic response, with positive effects seen with lower dosages, and inhibitory effects at higher dosages.³⁰ Treatment with a total dose of 20 J/cm² divided between 638 and 830 nm light produced better outcomes than using either wavelength alone, suggesting a possible effect of both superficial and deep treatments.³³ However, two studies with infrared laser treatment using higher dosages had reduced stimulation³⁴ or inhibitory effects.³⁵ In one study of rats with circular wounds receiving a single 830-nm laser treatment with 1.3 J/cm², more rapid epithelialization and wound contraction resulted compared with 3 J/cm², although both treatments resulted in stimulation of healing compared with controls, and there was no difference between the treated groups by day 14.³⁴ In a mouse wound model, 18 J/cm² of 980-nm laser treatment had a beneficial effect on wound healing, whereas 36 J/cm² was too aggressive and resulted in decreased healing.³⁵

Laser therapy may also be beneficial for difficult wounds in metabolically compromised patients. Laser

photostimulation accelerated wound healing in diabetic rats in one study.¹ Diabetes was induced in male rats by streptozotocin injection and two 6-mm diameter circular wounds were created on either side of the spine. The left wound of each animal was treated with a 632.8 nm HeNe laser at a dose of 1 J/cm² 5 days per week until the wounds closed (3 weeks). There was a marginal increase of biomechanical properties of the laser-treated wounds, including an increase in maximum load (16%), stress (16%), strain (27%), energy absorption (47%), and toughness (84%) compared with control wounds. The amount of total collagen was significantly increased in laser-treated wounds. It was concluded that laser photostimulation promoted tissue repair by accelerating collagen production and promoting overall connective tissue stability in healing wounds of diabetic rats. HeNe lasers also improved wound healing by increasing collagen synthesis when corticosteroids or nonsteroidal antiinflammatory agents were administered to rats with surgical abdominal wounds.³⁶ Laser treatment significantly increased collagen synthesis.

One study reported 100 clinical cases with healing wounds treated with LLLT.³⁷ There was a marked increase in collagen formation, increased vasodilation, and accelerated DNA synthesis. This researcher recommended 1 J/cm² of laser treatment.

A statistical metaanalysis was performed to determine the overall treatment effects of laser phototherapy on tissue repair.¹⁸ After performing a literature search, the effectiveness of laser treatment was calculated from each study using standard procedures. Thirty-four peer-reviewed papers met the inclusion criteria for tissue repair. There was a positive effect of laser phototherapy on tissue repair. Collagen formation, rate of healing, tensile stress and strength, time needed for wound closure, number and rate of degranulation of mast cells, and flap survival were improved with laser therapy. Laser treatment with a wavelength of 632.8 nm had the greatest effect, and 780 nm had the least. This metaanalysis concluded that laser therapy was an effective treatment for tissue repair.

Another study reviewed the literature regarding the in vitro and in vivo effects of low-intensity laser therapy on the wound healing process, especially in diabetic patients.³⁸ Although many of the in vivo studies lacked specific information on dosimetric data and appropriate controls, the data from appropriately designed studies indicated that LLLT should be considered as an adjunctive adjuvant therapy for refractory wound-healing disorders, including those experienced by diabetic patients.

However, results from other controlled and blinded studies have been less clear regarding the efficacy of LLLT for the treatment of wounds.¹¹ Studies in laboratory animals have suggested that LLLT may improve healing during the early stages of wound healing, but the effect may not result in improved total healing time.^{39,40} Another large review paper did not find unequivocal evidence

that LLLT was beneficial for the treatment of wound healing.⁴¹

Bone and Cartilage Effects

Bone and cartilage may also be affected by laser treatment. In one study of bone healing, rats received a defect in a femur.⁴² Rats were treated for either 12 sessions (4.8 J/cm² per session, 28-day follow up) or three sessions (4.8 J/cm² per session, 7-day follow up) with 40 mW 830 nm laser light. Treatments were applied three times per week, and two other groups served as untreated controls. Rats were sacrificed on either day 7 or 28 after surgery. Although there were significant differences between treated and control animals regarding the area of mineralized bone at 7 days, there were no differences at 28 days. The authors suggested that LLLT may have some effect on early bone repair.

Another study of bone healing in rats compared ultrasound with laser.⁴³ A GaAlAs laser (780 nm) was applied using 30-mW power and a dose of 112.5 J/cm². The ultrasound group was treated with 1.5 MHz, at 30 mW/cm². Both groups received 12 total treatments, with 5 treatments administered per week. Bones were harvested on day 20. Maximum load at failure was greatest in the laser-treated group. Ultrasound resulted in promotion of bone resorption at the osteotomy site, whereas laser caused bone formation in comparison with control osteotomies.

The use of GaAlAs laser was compared with bone morphogenetic protein (BMP) and bovine organic bone graft in a rat femoral bone healing model.⁴⁴ Rats were assigned to four groups: control; laser treatment; BMPs plus organic bovine bone graft; and BMPs plus organic bovine bone graft plus laser treatment. The laser-irradiated groups received treatments every 48 hours for seven treatments, beginning immediately after the surgical procedure. The laser therapy (830 nm, 40 mW) consisted of 16 J/cm² per session divided equally over four points (4 J/cm² each) around the defect. The subjects were sacrificed after 15, 21, and 30 days. There was increased deposition of collagen at 15 and 21 days, as well as increased bone trabeculae at the end of the experimental period in the irradiated animals versus the nonirradiated controls. The greatest healing was seen in the group treated with organic bovine bone graft, BMP, and laser treatment.

LLLT has also been investigated as an adjunctive treatment for infections of the musculoskeletal system. In one study, the effect of various doses of 808 nm (100 mW continuous laser, at doses of 7.64 J/cm², 15.29 J/cm², and 22.93 J/cm²) laser therapy was evaluated in induced chronic osteomyelitis of rat tibias, created with methicillin-resistant *Staphylococcus aureus*.⁴⁵ Rats had either surgical debridement, surgical debridement plus laser therapy, or no treatment. Infection levels decreased by 37%, 67%, 81%, and 93% in groups treated by debridement only, or debridement plus laser at 7.64 J/cm², 15.29 J/cm², or 22.93 J/cm², respectively, compared with the nontreated control group. The authors

concluded that laser therapy may be an adjunctive treatment for the management of osteomyelitis. Laser has also been used to create a laser shockwave to clear biofilm from medical devices *in vitro*.⁴⁶ Bacteria surrounding medical devices, such as stents and orthopedic implants, may make a biofilm coating that prevents penetration of antibiotics. In this study, the biofilm was disrupted by a pulsed Nd:YAG laser that produced laser-generated shockwaves, making the bacteria susceptible to conventional treatment.

A more complete review of the effects of laser therapy for bone repair has been published.⁴⁷ Most of the research regarding bone healing has been performed in cell culture or in rodent models. However, the authors acknowledge that more study on the laser properties, wavelength, and energy dosage is needed, along with improved study design.

The safety of laser therapy on epiphyseal cartilage has been evaluated in rats.⁴⁸ Young rats had either no laser irradiation, 5 J/cm², or 15 J/cm² of 830 nm GaAlAs laser application to the proximal tibial epiphyseal cartilage every other day for a total of 10 sessions. Laser irradiation increased epiphyseal cartilage thickness and the number of chondrocytes, but these effects were not great enough to increase bone length.

Another study evaluated osteochondral lesions of the knee treated intraoperatively with laser therapy in rabbits.⁴⁹ Bilateral osteochondral lesions were created in the femoral medial condyles. All of the left lesions underwent immediate stimulation using a GaAlAs laser (780 nm), and the right knees were left untreated as a control group. After 24 weeks, the condyles were examined histomorphometrically. The condyle treated with laser had better cell morphologic findings and repair of osteocartilaginous tissue.

The effect of laser therapy on cartilage was further investigated in another study that evaluated whether intraoperative laser biostimulation can enhance healing of cartilaginous lesions of the knee in rabbits.⁵⁰ Bilateral chondral lesions were created in the medial femoral condyles. The lesion in the left knee of each animal was treated intraoperatively using a diode GaAlAs 780-nm laser (300 J/cm², 1 W, 300 Hz, 10 minutes), and the right knee was left untreated. Cartilage was then examined 2, 6, or 12 weeks after surgery. The rabbits receiving laser therapy had progressive filling with fibrous tissue of the cartilaginous lesion, whereas no changes were apparent in the untreated group.

Although laser therapy may have some benefit to treat cartilage injury, it may also have benefit to help maintain the health of cartilage during periods of disuse and immobilization. The influence of laser therapy (632.8 nm, HeNe, 13 J/cm², three times a week) on the articular cartilage of rabbit stifles immobilized for 13 weeks was examined in one study.⁵¹ The number of chondrocytes and depth of articular cartilage of the treated rabbits were significantly higher than those of the sham-treated group. The cartilage surface of the sham-treated group was rough and fibrillated, whereas the surface of the experimental group was

intermediate between a nonimmobilized control group and the sham-treated group. The authors of this study concluded that low-power HeNe laser irradiation reduced the adverse effects on the articular cartilage of rabbits immobilized for 13 weeks. Another study evaluated the use of 810 nm laser therapy on bone and cartilage during joint immobilization of rat knees.⁵² Three groups of rats received either 3.9 W/cm², 5.8 W/cm², or sham treatment. After 6 treatments over a 2-week period, tissues were harvested for testing. Results indicated that cartilage stiffness, assessed by indentation testing, was preserved in both laser groups.

Treatment of Arthritis with Low-Level Laser Therapy

There is much interest in treating various forms of arthritis with laser therapy, including osteoarthritis (OA) and various forms of inflammatory arthritis, such as rheumatoid arthritis. Models of various forms of arthritis have also been studied in animal models. Zymosan was injected into knee joints of rats to create inflammatory arthritis in one study.⁵³ LLLT (830 and 635 nm, 20 mW) or LED (628 nm) was applied to animals immediately and 1 and 2 hours after injection, using the same dose of 2.5 J/cm². A positive control group was pretreated with a dexamethasone injection into the knee 1 hour before zymosan injection. Laser treatment significantly reduced edema by 23%, vascular permeability by 24%, and pain by 59%, whereas LED treatment had no effect on any of the outcome parameters. Laser treatment may inhibit inflammation by reducing PGE₂ levels by inhibiting the COX-2 enzyme. Another inflammatory joint model also suggested that laser treatment may reduce inflammatory changes.⁵⁴ Hydroxyapatite and calcium pyrophosphate crystals injected into rat knee joints resulted in inflammatory changes in the non-laser-treated control animals, but HeNe laser (633 nm, 5 mW, 8 J/cm² daily for 5 days) reduced the intensity of the inflammatory changes.

Laser therapy has been used for the treatment of OA in people. The effect of laser therapy in OA of the knee was investigated in one double-blind study.⁵⁵ One treated group that received infrared laser (GaAlAs) and the other that received HeNe laser treatment were compared with a group that received placebo treatment. Patients were treated for 15 minutes twice daily for 10 days. Total dose for each session was 10.3 J for HeNe and 11.1 J for GaAlAs. The laser-treated groups were significantly less painful compared with the placebo groups, but there was no difference between the HeNe and the GaAlAs groups. The Disability Index Questionnaire also revealed an improvement in the laser groups. Patients receiving laser treatment had less pain for 2 months to 1 year after treatment.

Laser treatment was also performed on 20 human patients with OA of the knee, ranging from 42 to 60 years of age.³ All patients had previously received conservative treatment with poor results. The laser device used for this

treatment was a pulsed infrared diode laser, 810 nm wavelength, once per day for 5 consecutive days, followed by a 2-day rest interval. The total number of applications was 12 sessions. Laser treatment was performed on five periarthritic tender points, for 2 minutes each. Pain relief and functional ability were assessed using a numerical rating scale, self-assessment by the patient, index of severity for OA of the knee, and analgesic requirements for comfort. There was significant improvement in pain relief and quality of life in 70% of patients, compared with their previous status, but there was no significant changes in range of motion (ROM) of the knee. Although the authors of this study indicated that laser treatment was beneficial, there was not an untreated control group. Therefore the results should be interpreted with caution.

A double-blind randomized study was conducted in 90 patients with OA of the knee to evaluate a GaAs laser in combination with 30 minutes of exercise.⁵⁶ One group received 5 minutes of laser therapy, with 3 J delivered. Another group was treated for 3 minutes and received 2 J, and a third group received placebo laser and exercise. Patients received a total of 10 treatments and were studied for 14 weeks. Patients receiving laser treatment had significantly improved pain, function, and quality of life measures after treatment and had improved scores as compared with the placebo laser group.

The analgesic and microcirculatory effects of LLLT were evaluated in a recent study of people with knee OA.⁵⁷ Laser therapy (830 nm, continuous wave, 50 mW, 6 W per treatment site) or placebo treatments were administered twice weekly for 4 weeks. Pain, circumference, pressure sensitivity, and flexion were all improved in the laser treatment group, whereas the placebo group had no improvement in flexion or pain. Thermographic measurements showed a 0.5°C increase in temperature in the laser group, suggesting an increase in circulation.

A similar randomized, placebo-controlled study of 60 patients with knee OA indicated no significant improvement with 50 mW 830 nm GaAlAs LLLT at 3 weeks or 6 months.⁵⁸ In this study, patients received 3 or 1.5 J per painful joint, or placebo laser treatment, five times per week, with 10 total treatments.

Finally, a metaanalysis review was conducted on the efficacy of laser therapy on OA in people.⁵⁹ Seven trials were included, with 184 patients randomized to laser, and 161 patients to placebo groups, using a variety of lasers and treatment protocols. Treatment duration ranged from 4 to 12 weeks. Pain was assessed in four trials. The pooled estimate of three trials showed no effect on pain measured using a scale, and two demonstrated very beneficial effects with laser. In another trial, with no scale-based pain outcome, significantly more patients reported pain relief (yes/no) with laser. One study found knee ROM was significantly increased. Other outcomes of joint tenderness and strength were not significant. Lower dosages of laser

were found to be as effective as higher dosages for reducing pain and improving knee ROM. The authors concluded that for OA, the results are conflicting in different studies and may depend on the method of application and other features of laser application, including wavelength, treatment duration, dosage, and site of application over nerves instead of joints.

The efficacy of laser therapy for the treatment of rheumatoid arthritis in people has also been reviewed.⁶⁰ Five placebo-controlled trials evaluating 222 patients met inclusion criteria of the review. Treatments were typically administered two to three times per week for 4 weeks. Compared with a control group, laser therapy reduced pain by 1.10 points on a 10-point visual analogue scale (VAS) relative to placebo, reduced morning stiffness duration by 27.5 minutes, and increased tip to palm flexibility by 1.3 cm. Other outcomes such as functional assessment, ROM, and local swelling did not differ between groups. There were no differences based on laser therapy dosage, wavelength, site of application or treatment length. The authors concluded that laser therapy could be considered for short-term treatment for relief of pain and morning stiffness for patients with rheumatoid arthritis. Further evaluation of wavelength, treatment duration, dosage, and site of application over nerves instead of joints is needed. A recent study, however, generally showed no effect of laser therapy (GaAlAs laser, 785 nm, 3 J/cm², and mean power of 70 mW) on the hands of patients with rheumatoid arthritis.⁶¹

Treatment of Tendon and Ligament Conditions with Laser Therapy

Low-level and therapeutic laser has been recommended for the treatment of various tendon and ligament conditions, but the clinical efficacy of this treatment remains controversial. Laser therapy has been used in people for tendinopathies of the Achilles tendon, patella tendon, medial and lateral elbow tendons, and shoulder rotator cuff tendons.

The effect of GaAs laser therapy (904 nm, 4000 Hz, 0.5 to 1 J/cm²) for tendinitis was evaluated in 74 treated people and 68 people receiving placebo treatment in a randomized, blinded fashion.⁶² All patients received six treatments during a period of 3-4 weeks. Pain was estimated objectively using a pain threshold meter, and subjectively with a VAS before treatment, at the end of treatment, and 4 weeks after the end of treatment. Laser therapy generally had a significant, positive effect compared with placebo measured from the first assessment to 4 weeks after the end of treatment. Laser treatment appeared to be most effective for acute tendinitis.

Another study evaluated the effect of laser therapy in seven people with bilateral Achilles tendinitis to see if laser treatment has an antiinflammatory effect.⁶³ Laser therapy (904 nm, 5.4 J per point, 20 mW/cm²) or placebo was applied to either Achilles tendon in random fashion. PGE₂ was reduced 75 to 105 minutes after laser therapy, and pain

pressure threshold values increased after laser therapy. The authors concluded that laser therapy reduces pain and inflammation in people with acute Achilles tendinitis.

A rat model of partial calcaneal tendon lesions also evaluated the effects of laser therapy.⁶⁴ Sixty rats were assigned to five groups: standard control (no lesion induced and no laser therapy); control (lesion induced, but no laser therapy); and groups 3, 4, and 5, which suffered the lesion and underwent laser therapy for 3, 5, and 7 days, respectively. Laser therapy was delivered with a GaAsAl laser (830 nm), power of 40 mW, power density of 1.4 W/cm² with continuous waveform, with a dose of 4 J/cm², and total energy of 0.12 J every day during treatment, always in the same period, for 3, 5, or 7 days starting on the day the lesion was produced. Control lesions had significantly less collagen fiber organization compared with the standard controls. There were no differences in collagen organization between group 4 and standard control. Groups 3 and 5 were similar to each other and had greater collagen fiber organization than the animals with control lesions. Laser therapy was effective in the improvement of collagen fibers organization of the calcaneal tendon after undergoing a partial lesion in this study, and early treatment for 5 days appeared to give the optimal response in this study.

Lateral epicondylitis can be a challenging condition to treat in people. A GaAs laser (904 nm, 12 mW, 70 Hz, 0.36 J/point with four or five points treated over the most painful area, two to three treatments per week for a total of 10 treatments) or placebo treatment was applied to people with lateral epicondylitis.⁶⁵ Laser treatment resulted in significant improvement in some objective parameters, but not subjective parameters, perhaps because the subjective scales were not sensitive enough to detect small differences.

One recent review and metaanalysis evaluated the use of low-level laser for tendinopathies in people.⁶⁶ The results of 25 studies were conflicting, with roughly half showing a positive effect and half being inconclusive or showing no effect. Positive results from studies of lateral epicondylitis and Achilles tendinopathy appear to be related to the dosage of laser applied. Studies suggesting efficacy of laser treatment used doses similar to those recommended by the World Association of Laser Therapy, which suggests doses of 1-8 J and power less than 100 mW/cm² for superficial tendons. For deeper tendons, such as the rotator cuff, power up to 600 mW/cm² and total dosages of 3-9 J have been recommended.

Laser therapy may also be useful for other forms of soft-tissue injury, such as ligament healing. In one study, 24 rats received surgical transection to their right medial collateral ligament and eight received sham operation.⁶⁷ After surgery, 16 received a single dose of GaAlAs laser to their transected ligament for 7.5 minutes or 15 minutes, and eight served as control with placebo laser, whereas the sham group did not receive any treatment. The ligaments were biomechanically tested either 3 or 6 weeks following

surgery. The ultimate tensile strength and stiffness of laser and sham groups were larger than controls. The laser and sham groups had improved stiffness from 3 to 6 weeks. It was concluded that a single dose of laser therapy improved the biomechanical properties of healing medial collateral ligaments 3 and 6 weeks after injury. However, a randomized clinical trial of laser therapy for the treatment of ankle sprains in people indicated that laser treatment was not effective.⁶⁸

Analgesia and Pain Management

Interest in the use of laser therapy to treat acute and chronic pain has increased in recent years. In fact, the World Health Organization's Committee of the Decade of the Bone and Joint has recently incorporated laser therapy into guidelines for the treatment of neck pain in people.⁶⁹ Knowledge of the effects of laser irradiation on pain has increased in recent years.⁶⁹ Nociceptors, which are A δ and C peripheral nerve fibers, lie just beneath the surface of the skin. Because these nerve fibers are relatively superficial, they are within the depths of laser stimulation. In addition, neurons that supply blood vessels and contribute to vasoconstriction and vasodilatation associated with inflammation are also within the depth of laser penetration. Therefore laser has the potential to influence pain perception by direct or indirect actions on nociceptors, as well as modulation of inflammation, which may contribute to pain. In fact, one study showed a reduction in postoperative pain and drug intake in people after immediate postoperative laser treatment to a surgical incision.⁷⁰ Laser therapy may suppress central sensitization and result in long-term depression of persistent pain with repeated application. The exact mechanisms by which laser therapy results in analgesia are unknown, but are the focus of active research. Although both nociceptors and motor nerves may be affected by laser irradiation, sensory nerves appear to be preferentially affected. It is likely that application of laser results in mainly inhibitory effects on aspects of nerve function, especially pain receptors.

The results of studies regarding pain management with the use of laser have been somewhat controversial. However, studies performed have resulted in the approval by the FDA of 635 nm low-level lasers for the management of chronic, minor pain, such as OA and muscle spasms. Laser therapy may have some analgesic effects by blocking pain transmission to the brain. Some studies have shown changes in the conduction latencies of the radial and median nerve after laser therapy,^{71,72} but others have shown no effect.⁷³ Laser treatment may also increase the release of endorphins and enkephalins, which may further provide analgesic benefits. Laser therapy has been used to stimulate muscle trigger points and acupuncture points, which may provide pain relief.⁷⁴

Although the precise mechanism by which laser therapy may provide analgesia is unknown, several studies have

investigated other possible mechanisms. One study evaluated the effects of diode laser irradiation of peripheral nerves.⁷⁵ The response was evaluated by monitoring neuronal discharges from the L5 dorsal nerve roots elicited by application of various stimuli to the hindpaw of rats, including brush, pinch, cold, heat stimulation, and chemical stimulation by injection of turpentine. Diode laser irradiation (830 nm, 40 mW, 3 min, continuous wave) of the saphenous nerve significantly inhibited neuronal discharges elicited by pinch, cold, heat, and chemical stimulation, but not discharges induced by brush stimulation. These data suggest that laser irradiation may selectively inhibit nociceptive neuronal activities.

Another study evaluated the effect of laser therapy on the head of rats.⁷⁶ Rats received various combinations of laser (0, 6.4, and 12 J/cm²) and naloxone (0, 5, and 10 mg/kg) prior to a hot plate test. Laser therapy (820 nm, pulsing) was applied to the rats' skulls. Hindpaw lick latencies (in seconds) in response to the hot plate test were recorded immediately, 30 minutes, and 24 hours after the administration of treatment. When animals were tested immediately following laser irradiation at 12 J/cm², significant analgesia resulted. Treatment with naloxone at either dose antagonized this effect, but naloxone produced no significant hyperalgesia when given alone. This suggests that opioid peptide mechanisms may mediate the analgesic action of laser therapy of the cranium.

Recently, the effects of laser irradiation on peripheral nerves and their potential effects on analgesia have been reviewed.⁷⁷ A total of 44 studies were included in this review. In 13 of 18 human studies, pulsed- or continuous-wave visible and continuous-wave infrared laser irradiation slowed nerve conduction velocity and reduced the amplitude of compound action potentials. In 26 animal studies, infrared laser irradiation suppressed evoked action potentials, pain-related neurotransmitters, such as substance P, and proinflammatory mediators, such as bradykinin. Disruption of axonal flow and microtubule arrays may be responsible for neural inhibition. The varicosities that form as a result of laser therapy inhibit axonal flow and block the transport of ATP, which is necessary to generate action potentials. The authors concluded that a range of inhibitory effects on peripheral nerves has been demonstrated, which may reduce acute pain by inhibiting peripheral nociceptors. In chronic pain, spinal cord changes induced by laser irradiation may result in depression of pain. Laser therapy seems to be most effective in reducing nerve conduction velocity when it is applied to several points over the nerve, causing an additive effect as compared with irradiation of a single point. In addition, continuous-wave laser therapy appears to be more effective than pulsed-wave application.

Laser acupuncture is another potential method of achieving pain control that has recently generated interest.⁷⁴ Laser acupuncture is the stimulation of traditional acupuncture points with low-level laser. Most animal models of laser acupuncture studies have shown no benefit over placebo

treatment, although the power used in those studies was generally very low. The results of many studies of laser acupuncture in people are hampered by inappropriate or poor description of laser application. Many of these studies have shown no positive benefits, but they used very low power. One study of laser-mediated acupuncture anesthesia (2.8-6 mW, 632.8 nm) for tooth extraction or minor facial surgery reported a 95% success rate. However, the characteristics determining the success rate were not defined.

A randomized, double-blind, placebo-controlled study of laser therapy in 90 subjects with chronic neck pain was conducted to determine the efficacy of 300 mW, 830 nm laser.⁴ Subjects were randomized to receive 14 treatments over 7 weeks with either active or sham laser to painful regions of the neck. Assessments were made at baseline, week 7, and week 12. Mean VAS pain scores improved by 2.7 in the treated group and worsened by 0.3 in the control group. However, the treated group began the study with higher mean VAS scores, despite randomization. Significant improvements were also seen in the laser therapy group compared with placebo for other total outcome measures, but some component portions of assessments did not differ significantly between the two groups. Laser therapy, at the parameters used in this study, was efficacious in providing pain relief for patients with chronic neck pain over a period of 3 months.

The results of laser therapy on acute and chronic back pain may differ from those obtained from treatment of neck pain. A randomized, double-blind, placebo-controlled study compared the effectiveness of laser therapy on pain and functional capacity in people with acute and chronic low back pain caused by lumbar disk herniation (LDH).⁷⁸ Patients were randomly allocated into four groups. Group 1 (acute LDH, n = 20) received hot pack plus laser therapy; group 2 (chronic LDH, n = 20) received hot pack plus laser therapy; group 3 (acute LDH, n = 20) received hot pack plus placebo laser therapy, and group 4 (chronic LDH, n = 20) received hot pack plus placebo laser therapy, for 15 sessions over 3 weeks. Pain, ROM, and disability were evaluated using standard techniques, such as VAS and global assessment scores, before and after 3 weeks of treatment. After treatment, there were significant improvements in pain severity, patients' and physicians' global assessment, ROM, and disability scores in all groups. However, no significant differences were detected between the four treatment groups with respect to all outcome parameters. There were no differences between laser and placebo laser treatments on pain severity and functional capacity in patients with acute and chronic low back pain caused by LDH.

A randomized, double-blind study of 100 patients with neck and shoulder pain indicated that 90% of the patients in the treated group had at least a 30% improvement in the degree of pain relief, as compared with only 14% of the patients in the placebo group.⁷⁹ Most patients had reduction of their pain immediately after treatment, and the improvement was typically maintained for 24 hours. A follow-up

study of another 100 patients indicated that 65% of the treated patients had improvement in their pain, but only 12% of untreated patients improved.⁷⁹

A metaanalysis evaluated the effect of laser therapy on pain relief.¹⁸ Nine papers met the inclusion criteria for pain control. The overall treatment effect for pain control was positive. Another review of laser therapy with location-specific doses for pain from chronic joint disorders suggested that some benefit may be derived from the use of lasers.⁸⁰ A literature search identified 88 randomized, controlled trials, of which 20 trials included patients with chronic joint disease. Laser therapy was applied within the suggested dose-range to the knee or temporomandibular joint capsule to reduce pain in chronic joint disorders. The results showed a mean difference in change of pain using a VAS by 45.6% in favor of laser therapy. Global status was also improved 33.4% more in patients in the laser therapy group. Although laser therapy appeared to reduce pain in patients with chronic joint diseases, the heterogeneity in patient samples, treatment procedures, and trial design calls for cautious interpretation of the results.

Peripheral Nerves and Spinal Cord

Peripheral nerve and spinal cord injuries can be devastating to patients of all species. Studies have been recently conducted which, along with the few known side effects of laser therapy, make this an exciting potential treatment for nerve and spinal cord injuries.

Initial studies evaluated the use of laser therapy on peripheral nerve regeneration in a crushed sciatic nerve model in rats. Most of the studies indicated that laser therapy was effective in promoting nerve recovery.⁸¹⁻⁸³ Laser therapy was applied directly to the nerve or transcutaneously. With direct stimulation, wavelengths of 540, 633, and 780 nm were found to be effective. Subsequent studies indicated that laser therapy applied to the nerve injury and the corresponding segments of the spinal cord with 633 nm laser at a dose of 10 J/cm² seems to improve the outcome. Although it appears that laser therapy is most effective when applied during the early posttraumatic period,⁸⁴ laser therapy may also be beneficial for chronic nerve conditions. A study of human patients with incomplete peripheral nerve or brachial plexus injuries present for 6 months to several years indicated that laser therapy resulted in progressive improvement of peripheral nerve function.⁸⁵

Studies of rat sciatic nerve injuries indicated that there was increased functional activity, decreased scar tissue formation, decreased degeneration of motor neurons, and increased axonal growth and myelination with laser therapy applied to the spinal cord immediately after wounding, and for 30 minutes daily for 21 days using 16 mW 632 nm HeNe laser.⁸⁶ This study suggested that laser therapy applied directly to the spinal cord may improve the recovery of corresponding peripheral nerve injuries.

Additional studies on laser therapy applied to surgically repaired peripheral nerves have also shown promise in studies of rats.⁸⁷⁻⁸⁹

Laser therapy appears to enhance axonal sprouting, resulting in accelerated healing. Laser therapy applied to the corresponding segments of the spinal cord also helps to reduce the degenerative changes in the neurons and helps induce proliferation of astrocytes and oligodendrocytes, which may result in an increased ability to produce myelin.⁸⁹ In addition, laser therapy may enhance the production of proteins and growth factors associated with axonal sprouting, such as growth-associated protein-43 and calcitonin gene-related peptide, and may stimulate the proliferation of Schwann cells.

In an initial study, laminectomy and transection of the spinal cord at T12-L1 were performed in 17 dogs.⁸⁵ An autograft of the sciatic nerve was implanted in the injured area. Ten dogs received laser therapy for 20 days, and the others did not. The 7 that did not receive laser therapy were paralyzed, whereas the 10 treated dogs stood between 7 and 9 weeks, and walked between 9 and 12 weeks. Treated dogs did not have prominent scar tissue, and there were new axons and blood vessels originating in the spinal tissue and extending into the graft. However, the study did not report the results of neurologic testing to know whether or not the dogs were spinal walking or had functional walking ability.

More recently, a prospective study was performed to determine if low-level laser therapy and surgery would result in earlier ambulation than surgery alone for intervertebral disk herniation.⁹⁰ Thirty-six dogs with acute paralysis or nonambulatory paraparesis were randomly assigned to a surgery or surgery plus laser group. A 5 × 200-mW 810-nm cluster array laser was used to deliver 25 W/cm² to the skin over the spinal segment associated with the hemilaminectomy and the two adjacent ones (one cranial and one caudal). However, the dose in joules per cm² was not clear. The laser array was applied to each area for 5 days or until dogs became ambulatory with paraparesis or ataxia. The time to achieve this level of recovery was significantly lower in the low-level laser therapy group (median 3-5 days) than the control group (median 14 days). The authors concluded that low-level laser therapy in combination with surgery decreases the time to ambulation in dogs with T3-L3 myelopathy secondary to intervertebral disk herniation.

The use of transcranial laser or LED therapy has recently been reviewed for the treatment of stroke, traumatic brain injury, and neurodegenerative diseases in people.⁹¹ There is some apparent transmission of near infrared light through the scalp and skull to stimulate the cortical surface of the brain. Because cortical neurons are rich in mitochondria, these cells may be responsive to light therapy. There may be upregulation of cytoprotective antioxidant enzymes, heat shock proteins, and antiapoptotic

proteins with light therapy, perhaps helping to preserve neurons afflicted with hypoxic or traumatic events.⁹¹ In addition, light therapy may increase neurogenesis, which has been demonstrated in rat models of stroke.^{92,93} In particular, there may be newly formed neuronal cells in the area ipsilateral to the lesion; however, there may be no new cells in the area of the lesion. Transcranial laser therapy may increase ATP in cortical neurons.⁹¹ In human stroke patients, transcranial laser therapy may improve outcome when applied approximately 18 hours after moderate stroke over the entire head.⁹⁴ Transcranial laser treatment has also been investigated in mild traumatic brain injury in mice and has beneficial effects. There are also cases of treatment of people with reduced cognitive function after brain injury in which cognition was improved after laser treatment.⁹¹

Application of Laser Therapy to Dogs

Before applying laser therapy to a patient, there are two fundamental attributes that must be established. First, the type of laser must be known, as well as the wavelength. The wavelengths of most lasers used for laser treatment are typically in the infrared or near-infrared range of 600-1000 nm. Wavelengths of low-power lasers commonly used are 632.8 nm (HeNe, gas) in the visible light range, 810 nm (GaAlAs, diode), and 904 nm (GaAs, diode) in the infrared region of the light spectrum. The wavelength is the prime determinant of tissue penetration. Lasers that do not penetrate as deeply (630 to 740 nm) are suitable for acupuncture point stimulation and wound healing but have not proved their clinical effectiveness with deep-seated musculoskeletal conditions. Infrared lasers (750 to 1500 nm) penetrate more deeply and are used to treat trigger points, ligaments, joint capsules, and intraarticular structures. The output power (watts or milliwatts) must also be known. Based on this, and the condition to be treated, the dose of laser light (J/cm^2) is determined. For most conditions, a dose of 1 to 8 J/cm^2 is typically applied. The time the laser must be applied to an area to deliver the dose must be calculated. For example, if a 904-nm laser with a maximum output power of 250 mW is used, it will take 4 seconds to deliver 1 J.

$$0.250 \text{ W} = 1 \text{ J} / X \text{ seconds}$$

$$(0.250 \text{ W})(X \text{ seconds}) = 1 \text{ J}$$

$$X \text{ seconds} = 1 \text{ J} / 0.250 \text{ W}$$

$$X = 4 \text{ seconds}$$

With this particular laser, it will be necessary to hold the laser on one point for 4 seconds to deliver 1 J of energy. With a 1-W laser, this takes 1 second. It is crucial to always

understand what the particular laser unit possesses with regard to power and wavelengths, and the dose, in J/cm^2 , a condition requires. From there, it should be easy to determine how much time is needed for laser application. Some laser units compute this with preprogrammed software, whereas others require the therapist to calculate how many joules are needed and how many joules are emitted per second. Some units may take 1 second or less to deliver 1 J, whereas other units may take up to 20 seconds to deliver 1 J.

The greater the power density and longer the wavelength, the deeper the penetration is through tissues. Unfortunately, the optimal wavelengths, intensities, and dosages have not been adequately studied in dogs, and information in people is difficult to interpret because of different conditions and treatment regimens. More laser dosage is not necessarily better, because overdosing may retard the desired effect. A common mistake in using laser therapy, especially in wound healing, is to experience a positive result and automatically assume more will be better. The healing process can only progress at a certain rate.

Laser therapy is generally administered with a hand-held probe, with a small beam area that is useful to treat small surfaces; other lasers have several beam areas in the same unit to treat larger areas (Figure 21-9). Laser energy may be applied with the laser probe in contact with the skin, which eliminates reflection from the skin and minimizes beam divergence, or with the probe not held in contact. With the noncontact method, it is necessary to hold the probe perpendicular to the treatment area to minimize wave reflection and beam divergence. Noncontact application is recommended for wound treatment. The appropriate dosage may be applied to larger areas by administering the calculated dose to each individual site in a grid fashion, or by slowly moving the probe over the entire surface, being certain to evenly distribute the energy to each site. In any case, the probe should be held perpendicular to the skin. A coupling medium is not necessary, as in ultrasound, because the laser beam is not attenuated by air.

To maximize laser application, the hair should be clipped because 50% to 99% of the light may be absorbed by hair. Little is known about the transmission of laser light to deeper tissues in darker dogs, but HeNe laser energy is likely to be absorbed because of the pigment. It has been recommended that the dose be increased by 25% in treatment areas with dark pigmentation. Any iodine or povidone iodine should be washed off the area. Any topical medications, especially corticosteroids and other photosensitizing agents, should be removed. The therapist should wear protective eye wear, because damage may occur to the retina if the laser shines into the eyes, and the patient's eyes should also be protected (Box 21-3).



Figure 21-9 Laser therapy is generally administered with a hand-held probe, with a small beam area that is useful to treat small surfaces.

Box 21-3 Key Points for Laser Usage in Dogs

1. Clip hair.
2. Measure area to be treated (a playing card is 57 cm²).
3. Determine treatment dose.
4. Increase dose for dogs with dark skin by 25%.
5. Determine the number of J/cm², total joules, and the length of treatment time for laser application. If treating an area the size of a playing card with 10 J/cm², the total treatment is 600 J. If using a 10-W laser, the treatment time is 60 seconds.
6. Place safety goggles on all in the immediate area.
7. Hold laser perpendicular to the skin (direct contact minimizes laser light reflection; noncontact application is recommended for wounds).
8. Apply laser treatment using an overlapping grid technique to be certain that the treatment is applied equally to all areas.

Application for Various Conditions in Dogs

Laser therapy has been used for the treatment of wounds; pain; inflammation; OA; and muscle, ligament, and tendon injuries. In general, Class IV laser energy is applied in an overlapping grid or fanning fashion, always moving the laser to avoid heating the area. Other lasers may be applied

using a point-to-point application of a dose per unit area. One can imagine the area covered by a grid with square centimeters, and the laser treatment is given in the squares. The following doses have been suggested for treating various conditions. However, there is currently little or no research available regarding dose or frequency of treatment in dogs for different conditions; the information is extrapolated from research in other species and anecdotal information and should be used with caution (Box 21-4).

Open Wounds

Clinically, laser therapy has been applied to both traumatic and surgical wounds, lick granulomas, abrasions, and skin conditions associated with allergies. The causes and sources of lick granulomas should be treated appropriately to complement the laser treatments. Laser therapy assists with the reduction of inflammation and wound healing, as indicated previously. Pain should also be considered in wound management as well, and laser therapy may be helpful to provide some pain control. Traumatic wounds also need to be treated appropriately with regard to infections, wound dressings, and other related issues. Debriding and cleaning should be performed before laser therapy is performed. When treating wounds, it is beneficial to document the size of the wound with photographs and measurements. The wound should be measured at each treatment session to document progress or lack of progress. Corresponding medications and bandage changes should be documented as well.

Traumatic wounds, such as degloving injuries, are understandably very painful and the animal may not tolerate direct contact. The laser head should be held above the wound to avoid contamination of the wound. The head should be cleaned thoroughly with rubbing alcohol prior to and after treatment. Most laser heads have a small lip or surface on them, preventing direct contact with the lens and the wound. Laser treatment is applied directly to the wound at a treatment dosage of 2 to 8 J/cm². Acute wounds may be treated with a lower dosage daily for the first 7 to 10 days. The entire wound area should be treated, including the periphery of the wound.

Postsurgical Wounds

Postsurgical wounds may be treated immediately after the surgical procedure to assist with healing. The postsurgical wounds appropriate for laser are not limited to orthopedic and neurologic cases. General postsurgical wounds, including spays, other abdominal surgeries, dewclaw removals, benign soft tissue removals, and other soft tissue surgeries, may respond to laser therapy. Laser treatments assist with cell proliferation, antiinflammation, and DNA synthesis. Because of this, surgical wounds following removal of neoplasia should not be treated with laser.

Box 21-4 Dose Guidelines for Various Uses in Dogs

Analgesic Effect

- Muscle pain: For acute pain apply 2 to 4 J/cm²; for chronic pain, apply 4 to 8 J/cm².
- Joint pain: Apply 4 to 6 J/cm² for acute pain, and 4 to 8 J/cm² for chronic pain.

Antiinflammatory Effect

- Acute and subacute: Apply 1 to 6 J/cm².
- Chronic: Apply 4 to 8 J/cm².

Open Wounds

- Acute wounds: Apply 2-6 J/cm² sid for 7-10 days.
- Chronic wounds: Apply 2-8 J/cm² sid.
- Laser head should not be applied directly to the wound.
- Clean laser head before and after treatment.
- Do not apply laser to wounds following removal of neoplasia.

Postsurgical Wounds

- A daily dose of 1-3 J/cm² is recommended for the first 7 to 10 days if possible, followed by a 1- to 2-day break, continued until the wound is healed. If daily treatment is not possible, then treatment three times per week may be performed.

Lick Granulomas

- Administer 1-3 J/cm², directly over the entire granuloma and at least 1 cm from the periphery. Depending on the size of the granuloma, the wound should be treated as frequently as possible; daily to a few times a week is beneficial. The granuloma should be treated until the wound is healed and hair growth has resumed.

Osteoarthritis

- Administer 8-10 J/cm².
- Treat along the joint lines and surrounding area.
- Hip: Start treatment at the greater trochanter, then direct around the cranial, medial, and caudal surfaces of the hip in a circumferential pattern.

- Stifle: Start at the patella. The joint line may then be followed either medially or laterally to complete the full circumference.
- Hock and digits: Start below the point of the hock and circle around the hock while applying treatment. It is important that the anatomy of the tarsus be considered, with its distal extent at the tarsometatarsal junction, to be certain that the appropriate areas are treated. The calcaneal tendon may also be treated. Individual toes may be treated but it is recommended that a small laser apparatus be used.
- Shoulder: Start at the greater tubercle, continuing circumferentially around the joint. Because of the tissue depth in this area, a longer wavelength laser source is necessary. If the biceps tendon or other tendons are involved, the treatment should include the length of the tendon.
- Elbow: Start below the olecranon process and continue around the entire joint in a circumferential manner. Treating distally and proximally from the elbow also benefits the surrounding soft tissue.
- Carpus and digits: Start at the accessory carpal pad and then treat in a circumferential manner. Sweeping motions may be performed distally and proximally to cover the distal and proximal aspects of the joint as well as the surrounding soft tissue. The digits may be approached the same way, although it is sometimes difficult to move in a circumferential manner. The dorsal and palmar aspects of the digits may be treated.
- Cervical spine: Treat the entire cervical spine from the suboccipital area down to the upper thoracic region. The laser may be applied directly over the dorsal cervical spine, and then moved over the epaxial musculature on both sides. Do not apply the laser for a prolonged period over the carotid arteries.
- Thoracic and lumbar spine: Treat directly over the area as well as the epaxial and surrounding musculature.
- Tendon conditions (biceps tenosynovitis, supraspinatus tendonitis, patellar tendonitis, and other tendon inflammatory conditions): The length of the superficial aspect of the tendon should be treated along with the surrounding soft tissues.

sid, Once daily.

The laser apparatus should be held directly over the wound and either moved in a sweeping fashion or held directly over the wound. The recommended treatment schedule is 1-3 J/cm² daily for the first 7-10 days, with a 1- to 2-day break, then continuing laser treatment until the wound is healed. Realistically, if the animal is not in the hospital, it may be difficult to laser the wound daily. Therefore as frequent treatments as possible should be pursued, with a minimum of three times per week. Laser therapy should continue a few times per week until the wound has closed and healthy tissue has formed. Clinically, steps should be taken to avoid self-mutilation by the dog, such as application of an Elizabethan collar.

Lick Granulomas

Lick granulomas are a common problem seen in small animal practice and may be treated with laser therapy. The source of the lick granuloma must be initially identified and treated appropriately, and then the wound may be addressed. There are often underlying causes of the lick granuloma. The lick granuloma should be treated with 1-3 J/cm², directly over the granuloma and at least 1 cm from the periphery. Laser treatment contributes to wound healing, decreases inflammation, and also contributes anti-histamine effects. Depending on the size of the granuloma, the wound should be treated frequently. Daily to a few

times a week is beneficial. The granuloma should be treated until the wound is healed and hair growth has resumed. However, the underlying cause must be determined and treated, whether it is anxiety, neurologic, orthopedic, or the result of other causes.

Osteoarthritis

As in human medicine, OA is very common in dogs. The initial step when addressing the arthritic patient is a thorough assessment of ROM, weight-bearing status, girth measurements, palpation, overall function and owner perception of problems with home activities. A pain scale may be used to determine the level of pain. Weight-bearing status may be determined through dynamic or static force-plates, or scales. The overall function and goals of the owner should be determined in the evaluation, and progress using the various assessment tools should be monitored.

After the problem list and the goals have been determined, treatment with laser therapy may be implemented as part of a multimodal treatment program. The specific goals of laser therapy often include decreased pain, inflammation, and improved circulation. Arthritic joints should be thoroughly treated along the joint lines and surrounding area (Figure 21-10). Recommendations for arthritic joints have ranged from 4 J/cm² up to 30 J/cm², but more appropriate doses may be 8-10 J/cm². The apparatus may either be held directly over the area for the designated number of joules, or used in a sweeping fashion depending on the unit.

Hip

Canine hip dysplasia and resultant OA are very common. The coxofemoral joint should be completely treated. A good landmark to start treatment is the greater trochanter. The treatment should then be directed around the cranial, medial, and caudal surfaces of the hip. The treatment is directed in a circumferential pattern. Referred pain or

secondary pain should also be addressed. The lumbosacral area and the musculature of the hip are often painful because of compensations and may also be treated.

Stifle

Arthritic stifles are easily treated with therapeutic laser. A nice starting point is the patella. The joint line may then be followed either medially or laterally to complete the full circumference. The patella ligament may be treated proximally and distally. Cranial cruciate repairs or deficient stifles often have soreness in the distal hamstring muscles and patients may benefit from treating this area as well.

Hock and Digits

Hock and digit joints are often difficult to treat with modalities such as cryotherapy or moist heat because of their size and irregular contour. Laser therapy is an easy treatment for these areas. Arthritic conditions of the hock may be treated by beginning below the point of the hock and circling around the hock while applying treatment. It is important that the anatomy of the tarsus be considered, with its distal extent at the tarsometatarsal junction, to be certain that the appropriate areas are treated. The calcaneal tendon may also be treated. Individual toes may be treated, but it is recommended that a small laser apparatus be used. If only a large unit is available, it may be used in a sweeping motion over the affected digits.

Shoulder

The shoulder and affected areas are easily treated. A good starting point is the greater tubercle, with the laser continuing circumferentially around the joint. Because of the tissue depth in this area, a longer wavelength laser source is necessary. If the biceps tendon or other tendons are involved, the treatment should include the length of the tendon. The cervical region and the cervical strap muscles may benefit from treatment because of possible secondary compensations that may occur from shoulder issues and lameness.

Elbow

The elbow is a very common area for arthritic changes and may benefit from laser therapy. The laser should be directed around the entire joint in a circumferential manner. A good starting point is below the olecranon process; treatment should then proceed around the area. Treating distally and proximally from the elbow also benefits the surrounding soft tissue, in particular the triceps muscles. Secondary and tertiary soreness and pain may be noted in the shoulder and the carpal region, and they may benefit from laser treatment.

Carpus and Digits

The carpal joint usually responds well to laser therapy to help reduce the inflammation and pain associated with



Figure 21-10 Treatment of an arthritic joint with a therapeutic laser.

arthritic changes. Arthritic digits may also be aided by therapeutic effects of laser, in addition to sesamoiditis. The carpal joint should be treated in a circumferential manner. The accessory carpal pad is a good starting point; the laser should then be directed around the joint. Sweeping motions may be performed distally and proximally to cover the distal and proximal aspects of the joint as well as the surrounding soft tissue. The digits may be approached in the same way, although it is sometimes difficult to move in a circumferential manner. The dorsal and palmar aspects of the digits may be treated, however.

Cervical Spine

The entire cervical spine should be treated from the suboccipital area down to the upper thoracic region. The laser may be applied directly over the dorsal cervical spine, and then moved over the epaxial musculature on both sides. Caution should be exercised to be certain that the laser is not applied for a prolonged period over the carotid arteries. The cervical musculature is thick and arthritic conditions of the cervical spine may benefit from the effects of laser.

Thoracic Spine

The lower thoracic spine is especially affected with arthritic changes and compensations. Treating directly over the area as well as the epaxial and surrounding musculature may be very beneficial.

Lumbar Spine

The lumbar spine, surrounding areas, and musculature may be treated the same as the thoracic spine.

Tendon Conditions

Biceps tenosynovitis, supraspinatus tendonitis, patellar tendonitis, and other inflammatory conditions involving tendons may be treated with laser therapy. The length of the superficial aspect of the tendon should be treated along with the surrounding soft tissues. The biceps tendon communicates with the shoulder joint capsule; therefore the shoulder joint capsule should be treated as well.

Precautions Regarding the Use of Laser Therapy

In general, the thermal effects of laser may injure tissues, especially the eyes, and in some cases, skin. Even small amounts of laser light can cause permanent damage to the retina.¹⁰ Because lasers have coherent light, even a small amount of laser light may be focused on the retina through the eye, and as little as 10°C increase in temperature may cause permanent damage to the retinal photoreceptor cells. The visible and near-infrared regions of the electromagnetic spectrum may penetrate into the eye, but because there is visible light, the blink reflex helps protect against



Figure 21-11 Protective eyewear should always be used when using a laser therapy.

Box 21-5 Precautions Regarding Laser Therapy

- Use protective eye wear.
- Do not direct laser treatment to the eye.
- Use caution with laser beam reflection from metal surfaces.
- Use caution with treatment in the presence of the following conditions:
 - Pregnancy
 - Open fontanels
 - Growth plates of immature animals
 - Malignancies
 - Photosensitive areas of the skin
- Dark colored skin, tattoos, and hair can absorb laser light and undergo excessive heating, resulting in discomfort.

damage. Infrared laser light is not visible, and therefore a blink reflex does not occur as with bright visible light. A person exposed to laser light in an eye may not experience any pain or discomfort until a pop is heard as a result of explosive boiling and damage to the retina.¹⁰ Caution must be used to select appropriate eyewear to be certain that the particular wavelength will be blocked and that an appropriate optical density is used to attenuate the beam power (Figure 21-11).

Other contraindications and precautions to laser therapy include pregnancy, treatment over open fontanels or growth plates of immature animals, over malignancies, and over photosensitive areas of the skin.⁵ Dark-colored skin, tattoos, and hair can absorb laser light and undergo excessive heating when a Class IV laser is applied, resulting in discomfort (Box 21-5).

Summary of Laser Therapy

Lasers are a potentially useful tool in veterinary rehabilitation. Although their use remains controversial, several studies have demonstrated benefit to using laser therapy.

Especially promising for their use in veterinary rehabilitation are studies showing preservation of cartilage properties with treatment, improvement in peripheral nerve injuries, and as a possible adjunct to managing pain, such as in patients with OA. Laser therapy also appears to have benefit in early wound healing. Laser therapy is noninvasive and there are no reported side effects when used properly.

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22 Applications of Extracorporeal Shockwave in Small Animal Rehabilitation

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Extracorporeal shockwave therapy (ESWT) has been in use since the 1970s, when its original application was to provide a noninvasive means to fragment uroliths, allowing for subsequent passage of the small fragments through the urinary tract.¹⁻³ Extracorporeal shockwaves have also been applied to the fragmentation of sialoliths, choledocholiths, and pancreatoliths.⁴ Applications to enhance bone healing began after observations of the effects of shockwaves on the pelvis during animal lithotripsy studies.⁵⁻⁷ These studies showed that a significant osteogenic response, specifically callous formation and bony remodeling, was elicited after exposure to extracorporeal shockwaves.

The use of ESWT has also been investigated for the treatment of human musculoskeletal diseases such as pain associated with plantar fasciitis, lateral epicondylitis, calcifying tendinitis, and femoral head necrosis, as well as for the treatment of delayed union and nonunion fractures.^{1,8,9} Integration into treatment of soft tissue disorders, such as treatment of pain and erectile dysfunction associated with Peyronie's disease or hypertension secondary to stroke, has also gained favor.^{10,11}

The use of ESWT in equine musculoskeletal disorders includes insertion desmopathies, bone spavin, tendon and ligamentous calcification, navicular disease, exostoses, fractures and microfractures, back pain, and osteoarthritis (OA) of tarsometatarsal and distal intertarsal joints. Extracorporeal shockwaves have been used in dogs as a part of the treatment regimen for hypertrophic nonunions, tendinitis, spondylosis, and OA^{4-6,12} (Figure 22-1).

How Shockwaves Are Generated

Extracorporeal shockwaves are acoustic waves of high pressure and velocity produced outside of the body. These waves are characterized by high-amplitude acoustic pressures (20-100 megapascals [MPa]) with a short build-up time of approximately 5-10 nanoseconds, an exponential decay to baseline to a negative deflection of approximately 10 MPa, and a wave cycle time of 300 nanoseconds (Figure 22-2).^{2,6,13,14} These pressure waves differ from ultrasound waves because of their lower frequency, minimal tissue absorption, and lack of a thermal effect.²

Focused generators produce pressure waves in a fluid medium when electrical energy is converted to mechanical energy. These waves may be focused by reflection of the pressure waves within the generator to the focal point, which is at the treatment site (Figure 22-3).^{6,13} The focused waves behave as sound waves in the tissues, releasing energy when tissue density changes are encountered (i.e., ligament to bone interface).^{3,6} At the interface of density change, part of the wave is transmitted while the other part is reflected.⁷ The greater the change in tissue density, the greater the energy released.

Three types of focused shockwave generators exist: electrohydraulic, electromagnetic, and piezoelectric. Piezoelectric generators use crystals that expand and deform when stimulated by high-voltage electricity, initiating a pressure wave.^{2,7,13} The electromagnetic system consists of coils arranged in such a way as to generate opposite magnetic fields when high voltage electricity is applied.^{2,7} When a high current pulse is released through the coil, a strong magnetic field is generated, creating a high current in the opposing metal membrane. The membranes are forced away from each other as a result of the opposing currents, causing compression of the surrounding fluid medium, resulting in a shockwave.^{2,7,13} Electrohydraulic generators use a high-voltage spark gap, causing plasma bubble generation. Expansion of the bubble results in sonic pulse propagation and subsequent collapse results in a reverse pulse, manifesting a shockwave (Figure 22-4).^{2,7,13}

Delivery of Energy

Delivery of energy to tissue requires the use of coupling gel in conjunction with a percussive handheld applicator. Energy can be focused via spherical arrangement of crystals, spherical generator, or acoustic lenses. Focusing energy results in greater tissue depth penetration, up to 110 cm, with a rapid decline of energy at the periphery, allowing an intense amount of energy delivery to a relatively small area.^{6,13} Eighty percent of the maximum shockwave pressure resides in the focal area. Hence, its maximum value is in the focus with a decrease in pressure at the periphery of the focal area.⁴ Piezoelectric generators have a small focal volume with high-energy flux values and low overall energy



Figure 22-1 A focused extracorporeal shockwave unit.

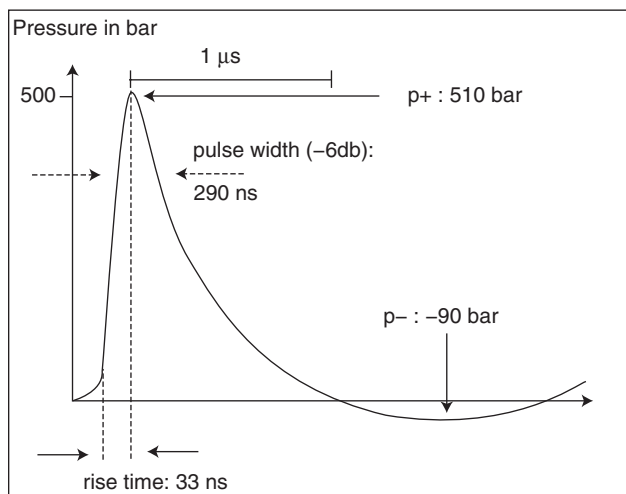


Figure 22-2 Pressure profile of an extracorporeal shock wave. Note the rapid increase to a relatively high pressure, followed by a rapid decline. (Courtesy of Pulsevet, Inc.)

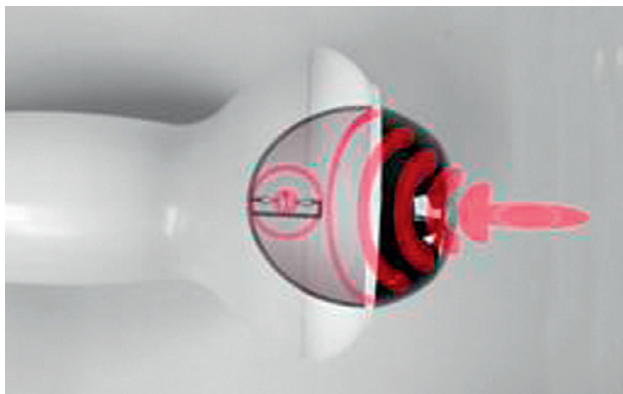


Figure 22-3 A trode used to apply extracorporeal shockwaves to a focused tissue depth. (Courtesy of Pulsevet, Inc.)

transfer. Electromagnetic generators also have a small focal volume and high-energy flux, but tend to be less concentrated than the piezoelectric system. Electrohydraulic generators produce the largest focal volumes with lower energy flux density and overall high energy transfer.¹³

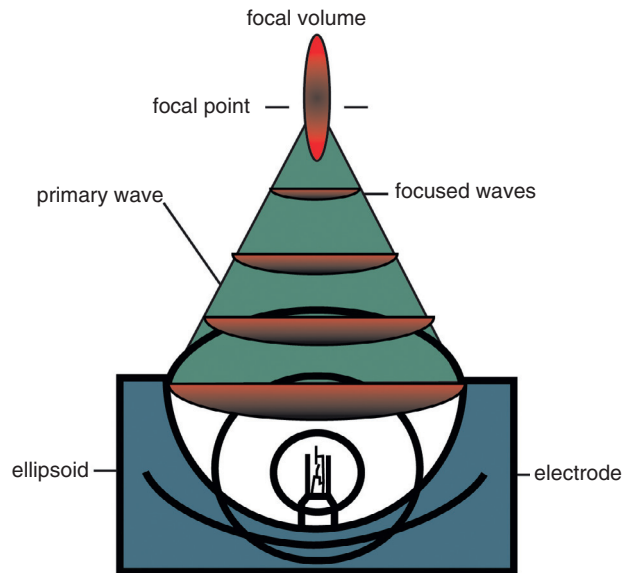


Figure 22-4 A spark gap focused shockwave generator. The shape of the parabola determines the depth of penetration of focused extracorporeal shockwaves. (Courtesy of Pulsevet, Inc.)

An alternative to focused extracorporeal shockwaves exists as a radial extracorporeal shockwave device, or a ballistic generator. Shockwaves from this device are generated pneumatically (mechanical concussion) and are propagated through tissues in a radial pattern. Radial pressure waves have lower energies, characterized by a slower rise time, and a negative component that is the same magnitude as the positive component.¹⁵ The energy applied by this device is greatest at the applicator-skin interface and decreases proportionally to the square of the distance from the applicator head, applying energy to a wider circumference with decreased tissue penetration.^{12,13,15-17} The rapid release of energy occurs over a relatively wide area with a rapid decrease in energy delivered to deeper tissues.^{5,6,12}

Research Regarding Extracorporeal Shockwave Therapy

General Biologic Effects

The primary effects of shockwaves are produced by the generation of mechanical forces. Secondary effects occur by indirect generation of mechanical forces through cavitation. Compression and tension generation occurs as the shockwave travels through the tissue. Varying amounts of reflection and transmission of energy occur at varying interfaces. In bone, for example, the amount of energy attenuation depends on the amount of cancellous bone, because attenuation of the shockwave increases exponentially with increased bone thickness.⁹ This is largely due to the various tissue interfaces; hence most of the energy from the shockwave is deposited in cancellous bone.^{2,6}

Cavitation bubbles are made during the tensile phase of the shockwave, and as size increases, a large amount of energy is delivered to the bubble. Cavitation bubbles formed secondary to shockwave-generated compression and tension forces have the potential for deposition of large amounts of energy after subsequent collapse, releasing high-energy water jets and high-temperature production.^{3,7} Cavitation at the surface interface is the mechanism by which fragmentation of uroliths occurs.³ Cavitation also leads to the production and release of free radicals, ultimately creating chemical reactions within the tissue. ESWT may also increase cell membrane permeability. The growth and collapse of cavitation bubbles, which may lead to surface deformation and damage is one potential mechanism of action for increasing membrane permeability.¹⁸

A dose-dependent cytotoxicity or decrease in cell viability is present with increased numbers of waves and energy density. Different cell populations differ in their sensitivity to shockwave energy. Bone marrow, for example, has a decreased sensitivity compared with malignant cell lines.¹⁹ Human fibroblasts have an increased sensitivity to higher numbers of shockwaves rather than the higher energy of shockwaves, decreasing the growth potential of fibroblasts while in suspension.²⁰ The reepithelialization rate of skin wounds in pigs is also affected by ESWT, and the effects depend on the energy level used.^{4,21}

Analgesia

The mechanism of action for induction of analgesia and tissue healing is poorly understood. Human patients tend to exhibit a bimodal analgesic effect, in which an initial decrease in pain 3–4 days after initial treatment has been attributed to the direct effects of shockwaves on nociceptors and impaired synthesis of substance P.²² After the initial analgesic effects, a gradual recurrence of pain is noted with a subsequent gradual resolution of pain noted over 3–4 weeks that is attributed to angiogenesis and tissue matrix remodeling.²² Analgesic effects in humans appear to be more consistent in chronically affected patients rather than acutely affected patients.²³ Horses have also exhibited bimodal analgesia; however, the mechanism remains unknown.¹³

It has been hypothesized that analgesia is induced by hyperstimulation of nociceptors resulting in an intense sensory input to the periaqueductal gray area of the midbrain. This sensory input promotes descending inhibition and modulation of sensory stimulus transmission through the dorsal horns of the spinal cord, known as the “Gate Control Theory.”^{15,24,25} A second theory is that induction of cell damage from shockwave application prevents appropriate membrane potentials required for transmission of signals.¹⁵ It has been shown that shockwaves induce production of cytokines and growth factors, endothelial nitric oxide synthase (eNOS) and up-regulation of bone morphogenetic protein (BMP) expression.^{26–28} As a result of cytokine and growth factor release, a decline in inflammation and swelling, with short-term analgesia, is noted.

Decreases in calcitonin gene-related peptide (CGRP) in free nerve endings and dorsal root ganglia (DRG) of rats after application of ESWT to the plantar skin of the hindlimb support the theory that altered nerve function of the locally treated area may be the source of analgesic effects.^{29,30} Substance P gained interest as a mode of analgesia, as an increase in basal concentration was noted 24 hours after ESWT application to the femur of a rabbit, followed by a gradual decrease in basal secretion over 6 weeks. This decrease was attributed to a potential degeneration of nerve endings in the treated area.³¹ Later studies by Haake and colleagues,^{32–34} did not find evidence of ESWT influence on spinal antinociceptive opioid mechanisms or on nonopioid mechanisms (substance P or CGRP). ESWT did not appear to activate peripheral nerves to induce analgesia, as evidenced by a lack of increase in c-Fos protein or its gene expression.³² Local desensitization and subsequent reinnervation has been theorized by Murata and colleagues,³⁵ in which an increase in activating transcription factor 3 and growth-associated phosphoprotein 43 was measured in rat DRG neurons after treatment with ESWT. Although a definitive mechanism of action and therefore treatment protocol for analgesia still remains to be elucidated, Takahashi and colleagues³⁶ described a cumulative effect of shockwaves on nerve fibers on the skin of rats, suggesting that multiple applications of low-energy shockwaves may provide longer-lasting antinociceptive effects.

McClure and colleagues²⁵ and Abed and colleagues³⁷ investigated analgesia induced by focused or radial shockwaves. Twenty-four horse limbs were assigned to receive control (sham treatment), ESWT, or radial pressure wave treatment (RPWT). Analysis of cutaneous analgesia revealed a substantial increase in cutaneous threshold to electrical stimuli for ESWT and RPWT at day 3. Analysis of neuropeptide concentrations, substance P, and CGRP did not reveal a significant difference between the groups.^{25,37} Sensory nerve conduction velocity was found to be statistically lower in nonfocused radial pressure waves when compared with control sensory nerve conduction. Disruption of the myelin sheath in medium- to large-diameter axons was noted under transmission electron microscopy, potentially creating a prolonged conduction velocity in radial pressure wave-treated nerves.¹⁵

There appear to be a number of potential mechanisms whereby ESWT exerts analgesic effects. The method of application, area of application, energy level, and frequency of application may have a role in determining whether the effect is directly related to pain pathway modification or stimulation of tissue healing.

Cartilage and Chondrocytes

There is tremendous interest in evaluating ESWT as a treatment for joint conditions, such as OA. Mueller and colleagues⁸ reported on the analgesic benefit of radial ESWT for hip OA in dogs, characterized by improved

ground reaction force difference between limbs 4 weeks following shockwave therapy and a potential long-term benefit of at least 3 months following treatment. Results of analgesia were similar to a study of canine stifle OA, in which three shockwave treatments (200 shockwaves were applied to each of four sites using a 20-mm focused depth applicator, followed by 175 shocks to each of the same four sites using a 5-mm focused depth applicator, at an energy flux density of 0.14 mJ/mm^2 , for a total of 1500 shocks) were administered 3 weeks apart. An improvement in peak vertical force and vertical impulse was evident at 21 days and continued to improve to the end of the study, at 98 days. Despite these improvements, the changes were not statistically significant from control groups perhaps because of small numbers of animals in each group.¹ In contrast, Francis and colleagues³⁸ reported significant improvement in ground reaction forces of dogs with elbow or hip OA when compared with control groups. A follow-up study of another group of dogs with elbow OA confirmed these findings and suggested that the amount of improvement with ESWT is similar to what would be expected when treating with an average nonsteroidal antiinflammatory drug (NSAID), despite the fact that the dogs were already receiving treatment for their arthritis, including NSAIDs.³⁹

Several studies have investigated how ESWT may affect the processes involved in OA. One study evaluated the effects of ESWT on the metabolism of human OA chondrocytes in vitro, especially the expression of β_1 integrin, interleukin (IL)-10, and tumor necrosis factor alpha (TNF- α).⁴⁰ Human articular cartilage was obtained from patients with knee OA undergoing total joint replacement. After isolation, chondrocytes underwent ESWT with different parameters of impulses, energy levels, and energy flux density. Chondrocyte intracellular IL-10 and TNF- α levels were evaluated. At baseline, osteoarthritic chondrocytes expressed significantly higher levels and IL-10 and TNF- α levels. Following ESWT, there was a significant decrease of IL-10 and TNF- α intracellular levels. IL-10 levels decreased with any number of impulses and energy levels, whereas TNF- α decreased at middle energies. ESWT significantly reduced the nitric oxide level in the synovial cavity of knee joints in a rabbit anterior cruciate ligament (ACL) transection model as compared with untreated rabbits with cruciate ligament transection.⁴¹ Chondrocyte apoptosis and the severity of cartilage lesions were also less in rabbits with OA treated with ESWT. The authors concluded that ESWT may be a useful treatment for knee OA because of the reduced progression of OA in rabbits. Although promising, further studies are necessary to determine if ESWT may modulate the changes seen with OA.

Another study evaluated the effects of ESWT on the development of OA in a rat ACL transection OA model.⁴² OA of the knee was detected radiographically 12 weeks after study initiation in the ACL-transected group,

but very subtle changes were noticed in the control non-transected knees and the ACL-transected plus ESWT groups (single treatment with 800 shocks, 0.18 mJ/mm^2). The articular cartilage of ACL-transected rats showed significant increases in cartilage degradation and chondrocyte apoptosis and significant decreases in subchondral bone remodeling compared with the control and ACL-transected plus ESWT rats. In addition, the ACL-transected plus ESWT rats had significantly decreased cartilage degradation and increased chondrocyte activity comparable to non-transected controls. The authors concluded that ESWT may have chondroprotective effects and improved subchondral bone remodeling in this rat OA knee model. In fact, ESWT of the subchondral bone may result in greatly reduced progression of OA.⁴³ Another study by the same group also showed that ESWT reduced cartilage changes in a rat OA model, with the most beneficial effects seen 4 weeks after ESWT. The group receiving ESWT had significantly lower histologic changes, increased collagen type II in the articular cartilage, and increased vascular endothelial growth factor (VEGF), BMP-2, and osteocalcin in the subchondral bone compared with the untreated rats.⁴⁴ Studies in a horse OA model, however, indicated no effect on subchondral bone, although increases in the serum biochemical markers osteocalcin and C-terminal telopeptide indicated bone remodeling.⁴⁵

Clinical function and the pain of OA may also be modulated by ESWT. In a rat OA model, walking duration on a treadmill was significantly extended 4, 7, and 14 days after ESWT in rats with knee OA, but began decreasing by day 21.⁴⁶ Although OA resulted in increased expression of CGRP in DRG neurons innervating the knee as compared with normal rats, ESWT reduced CGRP-positive DRG neurons, suggesting that ESWT is a useful treatment for knee OA.

Effects of ESWT on articular cartilage are of some concern, however, because if disruption of the structure of cartilage occurs secondary to ESWT application, joint health may be compromised. Gross disruption of cartilage integrity was absent after radial shockwave application to equine third metacarpal or metatarsal bone.¹⁶ A dose-dependent increase in lactate dehydrogenase after radial shockwave application supported the thought of increased chondrocyte membrane permeability.¹⁶ Focused shockwave effects on articular cartilage have also been investigated and these studies indicated a lack of radiographic, histologic, and macroscopic changes within joint cartilage of growing rabbits.⁴⁷ In vitro investigation of effects of focused shockwaves on articular chondrocytes revealed a dose-dependent increase in cytotoxicity, as well as decreased proliferative potential at higher energy doses.⁴⁸ Another study evaluated a single bout of ESWT (1500 shockwaves of 0.5 mJ/mm^2) on the structural integrity of articular cartilage of rat femoral heads in vivo.⁴⁹ Although there were no changes in histologic scores after ESWT,

positive immunostaining for tenascin-C and chitinase 3-like protein 1 was found up to 10 weeks after ESWT in treated cartilage and collagen type II mRNA was increased 1 and 4 weeks after ESWT. Other changes included expansion of the rough-surfaced endoplasmic reticulum, detachment of the cell membrane, and necrotic chondrocytes. Because all of these changes have been associated with early OA, there is some concern that high-energy ESWT directly applied to cartilage might cause degenerative changes in hyaline cartilage.

Tendon and Ligaments

In collagenase-induced models of suspensory ligament desmitis, destruction of abnormal metaplastic tissue and increased numbers of collagen fibrils was noted on histologic examination, suggesting possible promotion of healing of desmitis with ESWT.^{2,50} Evaluation of the bone-tendon interface in dogs and rabbits has revealed increased neovascularization after treatment, as indicated by an increase in neovessels and angiogenic markers, such as vascular endothelial growth factor, eNOS, and proliferating cell nuclear antigen expression.^{51,52} In a collagenase-induced patellar tendinopathy model in rabbits, increased tensile strength, hydroxyproline concentration, and tenocyte numbers, in addition to increased neovascularization, were reported.⁵³

In a rat model of collagenase-induced Achilles tendinitis, low-energy ESWT induced repair coinciding with an increase in transforming growth factor beta₁ (TGF- β_1) and insulin-like growth factor I (IGF-I) concentrations with tenocyte proliferation, neovascularization and progressive tissue regeneration, whereas higher energies impaired DNA, extracellular matrix synthesis, and biomechanical characteristics of tendons.²⁶ Increases in TGF- β_1 expression were noted in early tendon repair stages and IGF-I expression was increased throughout the healing process, indicating that TGF- β_1 and IGF-I are important mediators in ESWT-stimulated cell proliferation and tissue regeneration within a healing tendon.^{26,50} An increase in metachromasia was also noted and may be an indication of increased extracellular matrix production by cells.^{50,54} Increases in glycosaminoglycan and protein synthesis have been seen as early as 3 hours after ESWT application, with an overall tendency toward anabolic metabolism. Acceleration of healing may therefore be attributed to an initiation of ESWT-induced stimulation to the tissue with a subsequent decrease in cellular metabolism over 6 weeks.⁵⁵

Increased edema and inflammatory cells in the paratenon of rabbits has been described and could indicate that although ESWT has the potential to treat various tendinopathies via induction of tissue regeneration, there is also the potential for harm of tendinous tissue with application of ESWT.⁵⁵⁻⁵⁷ Documented changes appear to be dose dependent, either by the number of impulses or higher energy flux densities, and may decrease the

mechanical strength, making these structures susceptible to tearing.⁵⁸ The amount of energy and number of pulses delivered should be taken into consideration, because tendons and ligaments appear to be more susceptible to damage from excess energy.⁵⁴⁻⁵⁷

Radial pressure wave application has been investigated for the treatment of various tendinopathies, including insertional tendinopathies, with treatment successes reported in up to 80% of treated patients.^{59,60} In a study evaluating shockwave therapy for the treatment of supraspinatus calcifying tendinopathy, one dog exhibited subjective and objective improvement in lameness 21 days after shockwave therapy.⁶¹ Another dog in the same study was bilaterally affected and improvement was noted at 28 and 49 days after treatment in the right limb but not in the left limb. In both cases, disruption of the calcified material was not apparent after treatment.

ESWT has also been evaluated for the treatment of patellar ligament desmitis in dogs after undergoing tibial plateau leveling osteotomy (TPLO).⁶² Dogs that had TPLO surgery were evaluated by radiographs and ultrasonography preoperatively and 4, 6, and 8 weeks after TPLO. At 4 and 6 weeks, treated dogs received 600 shocks with an energy level of 0.15 mJ/mm², with a focused depth of application of 5 mm on the patellar ligament. There was a significant difference in distal patellar ligament thickness between groups at 6 and 8 weeks postoperatively. No significant ultrasonographic differences were found.

Soft Tissue and Wound Healing

Soft tissue healing may be promoted as a result of neovascularization and an increase in vascularity of the area. Neovascularization enhances healing and decreases pain associated with injury and can be seen as early as 4 weeks after treatment.⁵⁴ In a dorsal skin random flap model of rodents, there was a significant reduction in the distal necrotic area of flaps and a significant increase in tissue perfusion with a single treatment of ESWT when compared with control skin flaps. These findings indicate that a single dose of ESWT may enhance flap survival by increasing flap tissue perfusion. Increased expression of VEGF and proliferating cell nuclear antigen expression within the flap ischemic area indicated induction of neovascularization and cellular proliferation, a proposed mechanism for enhanced flap survival. A reduction in leukocyte infiltration and suppressed TNF- α suggested reduction of the inflammatory response; alternatively, it may result from the decrease in necrotic tissue within the ischemic zone.⁶³

Schaden and colleagues⁶⁴ studied the effects of radial shockwaves for the treatment of soft tissue wounds in humans. Response to treatment, defined as 100% wound epithelialization, was seen in 75% of wounds, in which causal factors or site specificity of the wound did not influence success of treatment. Acute wounds were more likely to heal completely than chronic wounds and large chronic

wounds demonstrated suboptimal healing. Overall, the use of unfocused, low-energy ESWT on patients with acute and chronic soft tissue wounds was associated with complete closure of 75% of wounds.

Haupt and colleagues²¹ described the effects of ESWT on partial thickness wound healing in piglets. Wounds treated with low doses of shockwaves had significantly enhanced rates of epithelialization. On histopathologic examination, wounds treated with low-dose shockwaves had a larger number of dilated vessels and macrophages in the perivascular space. The newly formed epithelial layer was twice the thickness as that of control wounds and wounds treated with high-dose shockwaves. In irradiated wounds, the rate of healing was increased by 50% when low-dose shockwave treatment was used.

The reepithelialization rate is inhibited with high-dose shockwaves, with intermediate energy having no effect on reepithelialization. Low energies appear to have a stimulatory effect on reepithelialization rate.⁴

Bone Healing

Bone healing is thought to be enhanced by extracorporeal shockwaves,⁶⁵⁻⁶⁷ possibly as a result of induction of microfractures within the trabecular bone, resulting in release of osteoinductive growth factors and the formation of a fracture hematoma. Bone marrow stromal growth and differentiation of osteoprogenitor cells have been correlated with induction of TGF- β_1 and VEGF, indicating that recruitment of bone-marrow mesenchymal cells may have a primary role in ESWT promoted bone repair.⁶⁸⁻⁷¹ Increases in BMP-2, BMP-3, BMP-4, and BMP-7 were documented as early as 4 weeks following ESWT treatment in a segmental defect model in rats, suggesting a vital role of BMP in ESWT-induced cell proliferation and bone regeneration.⁶⁹⁻⁷¹ Investigation into transformation of the physical stimulus of ESWT into an intracellular mitogenic response suggests that ESWT induces mitogen-activated protein kinases, extracellular signal-regulated kinase, and p38 kinase, contributing to bone regeneration in a rat segmental bone defect.⁷² Growth factors in turn induce neovascularization, and recruit inducible osteoprogenitor cells, resulting in osteoblast proliferation and bone formation.^{24,70} Endothelial NO synthase and an increase in BMP expression have been associated with osteosynthesis in treated areas. In a rabbit model, treated femurs were characterized by increases in periosteal bone formation secondary to an earlier increase in periosteal and trabecular bone disruption.²⁷

Energy levels that are too great may have deleterious effects on bone, however. Studies conducted on rabbits reported soft tissue and medullary edema, decreased blood flow and bone metabolism, periosteal detachment, and even fracture of cortical bone, with an increase in frequency and severity as the energy density increased.^{73,74} Other findings of microdamage of cortical bone were noted

in racehorses in an ex vivo model. More specifically, focused shockwaves were associated with an increase in microcrack density and radial shockwaves were associated with an increase in microcrack length.⁷⁵ In a study evaluating the acute effects of ESWT on the third metacarpal and metatarsal bone, however, McClure and colleagues² discovered that evidence of cortical microfractures and soft tissue damage was absent. Scintigraphic, thermographic, and histologic assessment of ESWT effects on the fourth metatarsal bone in normal horses did not reveal evidence of damage in the treated area.^{76,77} Absence of microfractures is not an indication of normal mechanical properties, as accumulation of microscopic damage was not detected in canine femurs until 15% of the modulus of elasticity was lost, indicating that accumulation is threshold dependent.¹² McClure and colleagues² postulated that the mechanisms for stimulation of osteogenesis included bone marrow hypoxia, subperiosteal hemorrhage, increased regional blood flow and activation of osteogenic factors rather than a result of cortical microfractures.² Evaluation of osteons in this study revealed the potential of ESWT to provide full cortical thickness stimulation of osteonal remodeling.

In humans, treatment of nonunion fractures with noninvasive ESWT resulted in a 76% success rate, compared with a 79% success rate with surgical therapy. The success rate improved when there was greater biologic activity in the nonunion, with a success rate of 100% for hypertrophic nonunions and 36.4% success rate in atrophic nonunion fractures. Current theories for enhancement of fracture healing include increases in bone metabolism secondary to increases in bone growth factors such as TGF- β_1 , basic fibroblast growth factor, VEGF, and osteocalcin measured in a rabbit atrophic nonunion model.²⁸ In a radial nonunion model performed in dogs, 80% of the dogs exhibited a bridging callous 6 weeks after shockwave treatment; by 9 weeks there was narrowing of the fracture gap and an increase in the bridging callous, and finally complete bony union at 12 weeks following treatment.⁴ Persistent radiographic nonunion was present in four of five of the control dogs and an incomplete bone union was evident in the last control dog. Wang and colleagues⁷⁸ described the effects of ESWT on an acute fracture model on 16 tibiae of eight laboratory dogs. In this study, radiographic callous formation was not significantly increased from control specimens until 12 weeks following treatment, although measurements were consistently greater in the treated group by 4 weeks. Cortical bone at 12 weeks was histologically more dense, thicker, and heavier than control callous, signifying the potential advantage of treating acute fractures that have the potential of becoming nonunions. In addition, fractured bone treated with high-energy ESWT has increased bone mineral density, higher ash and calcium content, and higher peak load, peak stress, and elastic modulus than in untreated fractures.^{67,71,79,80} Therefore it can be assumed that ESWT treatment improves the quality

and mechanical qualities of bone during fracture repair. Shockwave therapy is associated with decreased incidence of disturbed fracture healing, and may be indicated for use in patients with multiple fractures, open fractures, animals with a concurrent systemic disease, or geriatric animals.²⁴

ESWT also appears to be an effective treatment for human patients with osteonecrosis of the femoral head. Wang and colleagues⁸¹ compared use of ESWT treatment versus core decompression and bone grafting. Patients treated with ESWT had significantly improved pain and Harris hip scores at each follow up. Overall, 79% of patients showed improvement after treatment with ESWT and a trend toward a decrease in size of the lesion at each follow-up interval. In the rabbit model of avascular necrosis of the femoral head, an increase in VEGF mRNA expression was noted as early as 1 week after shockwave treatment and was maintained for 8 weeks after treatment; VEGF protein expression peaked at 4 weeks and was maintained for 12 weeks.⁸² Increased neovascularization was noted within regions of new bone formation 12 weeks after initial ESWT treatment in these rabbits, leading to the conclusion that ESWT therapy may promote repair of subchondral bone in avascular necrosis of the femoral head in rabbits.

ESWT may also be useful in treating osteomyelitis. Chronic osteomyelitis was induced in rabbits by injecting *Staphylococcus aureus* into the proximal tibia in one study.⁸³ Focused ESWT was applied twice to the infected limbs 4 and 5 weeks after inoculation. Signs of bacterial spread were not detectable after ESWT. Systemic indicators of infection were also unchanged. However, histologic scores of osteomyelitis were significantly decreased in the ESWT group compared with the untreated control, although *S. aureus* was still detectable. This study suggests that ESWT may be beneficial as an adjunct treatment of chronic bone infections.

Cardiovascular

Recent investigation of the effects of ESWT for the treatment of cardiovascular diseases has yielded promising results. Low-energy ESWT effectively induced angiogenesis and improved myocardial perfusion and cardiac function in a porcine model.^{84,85} Improvement of myocardial perfusion and cardiac function was also investigated in humans with end-stage coronary artery disease and refractory angina pectoris without adverse effects.^{86,87} Uwatoku and colleagues⁸⁴ recently investigated the effects of ESWT treatment on left ventricular remodeling secondary to acute myocardial infarction. Acute myocardial infarction was induced in a porcine model and then treated with ESWT 3 days or 4 weeks after myocardial insult. Results indicated that treatment with ESWT in the acute phase resulted in improved left ventricular remodeling and increased capillary density in the infarcted myocardial border zone. Improved myocardial perfusion secondary to angiogenesis

was associated with improved left ventricular function. ESWT may theoretically be extended for use in vascular complications, such as venous and arterial thrombi dissolution or modification of arterial plaques, as visible histomorphologic changes were documented when used on normal human aortic specimens in vitro.⁸⁸ Facilitation of endothelial progenitor cells secondary to ESWT has been associated with increased relative blood flow recovery in chronic ischemia of the myocardium, as well as in the hindlimb.^{85,86,89} Neovascularization occurred via upregulation of VEGF as well as stromal cell-derived factor 1, a chemoattractant that plays an important role in hematopoietic stem cell homing and function.

Application of Extracorporeal Shockwave Treatment

Light anesthesia or heavy sedation of the veterinary patient is generally administered before applying focused shockwave treatment to improve patient comfort. In a study investigating the effects of radial shockwaves on the treatment of hip OA, 18 dogs were treated without the use of sedation or anesthesia without evidence of adverse effects related to pain.⁸ A study in human patients compared effects of ESWT application in the absence of local anesthetics for the treatment of chronic plantar fasciitis and found that the pain score was significantly less in patients without local anesthetic.⁹⁰ The hair should be clipped over the site of application because air attenuates the sound waves, making the treatment less efficacious (Figure 22-5). The use of ultrasound gel between the skin and applicator is also necessary to eliminate air. A thorough understanding of the anatomy of the area where shockwaves are to be applied is necessary. Application of ESWT to a joint should



Figure 22-5 Application of extracorporeal shock wave treatment to an elbow joint of a dog. Note that the hair has been clipped and ultrasound gel has been applied.

be directed at the insertion of the joint capsule rather than the articular cartilage of the joint. Treatment of the entire joint capsule insertion regions may be more efficacious. When applying ESWT, the applicator should be gently moved along the joint capsule insertion sites and mildly rotated to be certain that all areas are treated. When treating ligamentous or tendinous injuries, direction of the wave should be in a proximal to distal direction. The optimal energy density, number of shockwaves applied, and number of sites per application has yet to be elucidated and guidelines for application are generally provided by the manufacturer.⁴ In general, 500-1000 shocks are applied for most conditions. Wounds should be treated with low energy levels, whereas nonunion fractures and OA may be treated with higher energy levels. The optimal frequency of application is also uncertain, but generally two treatments 3-4 weeks apart are recommended for most conditions. Improvement of conditions such as OA may occur for several months after treatment.

Adverse Events and Precautions with Use of ESWT

Negative effects may be observed with application of high-energy shockwaves, a large number of shocks administered, or if application is focused on single areas of sensitive structures. Lower energy levels generally reduce the risk of deleterious effects. Caution is advised in young or developing animals, because directing shockwaves over an open physis has been implicated in the development of focal tibial growth plate dysplasia in rats.⁹¹ However, other studies have contradicted Yeaman's results, demonstrating that under human shockwave therapy conditions, damage to the tibial and femoral epiphyses was not evident in a rabbit model.^{92,93} Väterlein and colleagues⁴⁷ and Lüssenhop and colleagues⁹⁴ reported shockwave-induced intramedullary ossification with slight shortening of the treated leg in rabbits; however, this shortening was not clinically significant. Interestingly, cartilage has been found to have less detrimental effects associated with shockwave therapy, even with higher energy levels.²⁷ The explanation for this is unknown, but it has been postulated that the decreased vascularity of cartilage may decrease deleterious effects, as cavitation bubble formation requires free fluid to form before tissue damage takes place.²⁷

The most common side effects after shockwave therapy include petechiae and bruising. High-energy flux densities may be associated with small hematoma formation and local swelling at the application site.^{2,4,9} Patients may have discomfort after application, requiring the use of analgesic medications. Corticosteroids and NSAIDs are generally not used unless necessary. Interstitial damage of arterial walls can develop if they are within the focal zone, resulting in rupture, vasoconstriction, or increased permeability. Repetitive action potentials of nerves can result from

cavitation induction with ESWT. As a result, major arteries and nerves should be excluded from the focal zone.^{2,9} However, nerves and vessels appear to be less sensitive to the effects of ESWT as compared with other tissues.

Indications for Use

ESWT may be beneficial as an ancillary treatment for OA as part of the multimodal approach to therapy of this condition.⁹ Improved weight bearing and passive range of motion are similar to results expected with NSAID use. In patients that are unable to tolerate the administration of NSAIDs, extracorporeal shockwaves may serve as a useful, noninvasive alternative for treatment of osteoarthritic conditions. Anecdotally, treatment of conditions affecting the elbow, hip, or back may respond better than other joints. ESWT has been used for delayed or nonunion fractures. Tendinopathies and ligament conditions may also respond to ESWT. Wounds may also benefit from treatment with ESWT, especially if they are large wounds treated in the acute phase of wound healing. ESWT may provide some antiinflammatory treatment of soft tissues, recently attributed to a rapid enhancement of eNOS activity in human vein endothelial cells.⁹⁵

Contraindications to Shockwave Therapy

Extracorporeal shockwaves should not be used in dogs suspected of having immune-mediated joint disease, infectious arthritis, neoplastic disease, diskospondylitis, or unstable fractures, or in dogs with neurologic deficits. Lee and colleagues⁹⁶ studied the effects of various doses of ESWT on the spinal cord of rabbits. Neurologic deficits were not apparent after shockwave application; however, varying degrees of microscopic changes were present, including myelin damage and neuronal loss.⁹⁶ Use of ESWT near gas-filled cavities or an organ should be avoided, as the risk for damage is high because of potential high-energy release. Delivery of shockwaves over lung fields, brain, heart, major blood vessels, nerves, and a gravid uterus should be avoided. One study investigating the effect of shockwave application to pregnant rabbits at different gestational stages found that congestion and parenchymal hemorrhage was present in the liver, lung, and kidney of fetuses exposed in the early gestational period.⁹⁷

Avoiding application to sites with neoplasia or infection has been recommended because of the potential of inducing metastasis or septicemia from physical displacement of neoplastic cells or bacteria.^{2,9} In mouse melanoma and prostate tumor models, lithotripter shockwaves were applied to transplanted tumor cells on the hindlimb of each mouse. Enhanced metastasis was attributed to shockwave treatment secondary to cavitation.^{98,99} Conversely, enhancement of chemotherapeutic effects was observed when regional exposure to shockwaves was used in combination

with intravenous bleomycin and cisplatin. Proposed mechanisms for this finding include increased permeability as a result of hydrodynamic flow. Combination therapy resulted in inhibition of cancer cell proliferation and tumor growth in rats, and these effects appeared to be dose dependent.¹⁰⁰⁻¹⁰²

Summary

Extracorporeal shockwave therapy appears to have several applications in canine rehabilitation. Wound healing, osteoarthritis, delayed and nonunion fractures, and ligament and tendon conditions seem to be positively influenced by treatment with ESWT. Further information is needed regarding the energy level, number of shocks, and frequency of application to provide the optimal treatment for various conditions. In addition, additional research is required to further understand the mechanisms of action.

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